

(Un)Common™ – HMT

HUMIDITY • MOISTURE • TEMPERATURE

MAKE YOUR OWN GRAIN BIN MONITORING SYSTEM

Project Ideas and Programming Code by

The Computer Connectors LLC



WHY SPEND THOUSANDS WHEN YOU COULD SPEND JUST A FEW HUNDRED?!

Now Supporting 24 Grain Types!

- Know the condition of your grain as you place it into your bin
 - HMT – Humidity, Moisture, Temperature
- Know the condition of your grain for the duration of its storage in your bin
- Minimize spoilage
- Avoid losing money by over drying your grain
- Make more money by (legally) re-hydrating your grain to (near) industry standards
- (Hopefully) Minimize the need to enter your bin to remove/breakup spoilage
- (And, did we mention)... **Make More Money**

(Un)Common – HMT

Make Your Own Grain Bin Monitoring System

User's Manual

Getting Started

If you are reading this, you are probably a farmer.

Be proud of that!

Nobody feeds and fuels the world like you.

There are a lot of people who are secretly jealous of you. Yes - Seriously!

Why else would they spend so much time and effort trying to tear us down?

Ok, We know...

Nobody likes to read instruction manuals. We feel the same way.

In this case, however, we strongly recommend reading this manual in its entirety – at least once prior to beginning the build process.

Reading this manual will help you get a good feel for:

- Where you are starting from
- A realistic goal
- The tools you will need
- The project parts you will need to order
- The building process
- And finally, what to expect at the end of the build.

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THEORY OF OPERATION

Theory of Operation

Introduction

Throughout this document, you will find references to:

- Grain types
- Temperature
- Humidity
- Moisture
- Equilibrium
- Grain marketing standards
- Etc.

As a farmer, you may have any number of reasons for growing your crops. Yet, regardless of your reason(s), one of your goals is to preserve the quality of that crop for as long as necessary. That may be days, weeks, or even years.

This project is intended to help you with that goal. We cannot **prevent** quality degradation or even outright spoilage. However, we can provide you with information to help you minimize your risk.

With that said, **YOU** need to be the subject matter expert here! Chances are that you have been farming for a long time. **How have you minimized your crop storage losses in the past?**

Use our HMT device (the one you are building) as a supplement to and **NOT a replacement for the knowledge that you have built up over the years.**

Grain spoilage is, in many cases, the result of too much moisture (a.k.a water) in the grain. That moisture combined with oxygen and other components in the environment is all we need for bacteria, fungi, and even pests to thrive in, on, and around your grain. In theory then, we want to minimize the amount of moisture contained within your grain. Yet, we don't want to remove so much water as to then miss out on additional profits that would have resulted from the additional "water weight."

So, you can see... It's a [water] balancing act.

In general, you can "get away with" more moisture if you will be storing your grain for very short periods of time and/or in cold environments. For longer periods of time or in warm environments, you may have to sacrifice some of the additional "water weight" profit to ensure that you maintain your grains quality.

But, How Can You Know your grain's moisture content?!

What This Device Does

In very simple terms, our HMT device uses Temperature and Humidity sensors connected to a miniature computer. That computer reads those sensors and calculates the [estimated] amount of moisture (as a percentage) in your grain.

On the following pages, we will discuss this device in two different contexts: The “FREE” version and the “PRO” version. We will discuss the differences between the FREE and PRO versions in more depth later in this document.

Briefly, the first difference between the versions is the ability to read more sensors. With the FREE version, you have the ability to read two sensors. We will call the first sensor the “Weather” sensor. Its purpose is to sense the outdoor air conditions and report those values to the computer. The second sensor is installed inside the grain bin. Its purpose is to sense the temperature and humidity within [one portion of] the grain pile and report those values to the computer. The computer then uses either grain-specific tables or two mathematical formulas to compute and estimate your grain’s moisture content. The computer also uses those same tables/formulas to compute an “Equilibrium Moisture Content” of the outside air from the Weather sensor.

With the PRO version, you have the ability to add two additional grain sensors inside your bin. Advantage example: Know the condition of your grain in three locations (ex. Top, Middle, and Bottom).

Additionally, with the PRO version, the device has four “Outputs.” Two of the outputs are designed to power LED lights which can quickly and visually help you determine what the effect will be on your grain if you run your bin’s fan. The third and fourth outputs are designed to provide control signals to external, third-party equipment.

Equilibrium Moisture Content

You are probably wondering what an “Equilibrium Moisture Content” is. While the term sounds technical, think of it like this...

If you left your grain out in the field, its moisture content would fluctuate based upon the weather conditions. If the air is cool and humid, the grain may absorb moisture from the air. If the air is hot and dry, the grain may lose moisture to the air.

The same is true inside your grain bin. If you use the fan to pull outside cool and humid air through the grain in your bin, the grain inside your bin may take on moisture from that outside air. If you use the fan to pull outside hot and dry air through the grain in your bin, the grain may lose moisture to that exhausted outside air.

So, if we use the aforementioned tables and/or mathematical equations to calculate what moisture content your grain would “Equalize” to when out in the field, we will then also know what moisture content your grain would “Equalize” to when your bin fan pulls this outside air through the grain in your bin.

You just need to ask three questions:

- 1) What is the current moisture content of the grain in my bin?
- 2) Do I want to DRY the grain in my bin?
 - a. Turn the fan on when the outside air is hot and dry.
- 3) Do I want to REHYDRATE the grain in my bin?
 - a. Turn the fan on when the outside air is cool and humid.

This device helps you determine the answers to those questions.

Moisture Calculation

Decades ago, scientists realized the benefit of this moisture “balancing act.” They then proceeded to create mathematical formulas that now help us to estimate grain moisture content. Our HMT device uses a combination of lookup tables that are based upon those mathematical formulas or, depending upon the grain type, it uses the equations to compute an estimated grain moisture content.

NOTE: To learn more about “equilibrium moisture” and these two scientists, simply search the Internet for their names. There are literally thousands of documents that make great bedtime reading.

ALSO NOTE: The aforementioned equations ARE NOT and CANNOT be 100% accurate. The “equilibrium moisture” is not only dependent upon temperature and humidity. It is also dependent upon the physical makeup of the grain: Starches, cell structures, etc. Even differences in grain hybrids will cause differences in the equilibrium moisture content. Thus, the bottom line: Use the moisture values calculated by Henderson and Chung-Pfost as an ESTIMATE of the real moisture content in your grain.

Grain Rehydration Considerations and Cautions

A lot of hype has been made on agricultural TV shows about rehydrating your grain to make more money. **While rehydration is definitely possible, there are risks!** For instance, some grains expand in volume when rehydrated. **Can your bin tolerate that?** And, on a not-entirely unrelated note, we even had someone comment about snow and ice accumulating in their fan housing. An automated system to turn that fan on would have destroyed that fan.

It is for those (and other) reasons that we do not provide the circuitry necessary to “automatically” cycle your fan. The information that this device provides for you is intended exclusively as “additional information” that you (the subject matter expert) can use to make the necessary, real-time fan operation decisions.

We encourage you to read more about this subject. One excellent resource is from NDSU:

<https://www.ndsu.edu/agriculture/ag-hub/ag-topics/crop-production/drying-storage/considerations-when-conditioning-too-dry-soybeans>

List of Supported Grains

As of firmware version 5.4, we are proud to announce that the HMT device now supports 24 grains.

GRAIN	SHORTENED NAME (Displayed on LCD)	EMC SOURCE DATA (See Appendix B for additional source detail)
Barley	BARLEY	www.pami.ca
Canola	CANOLA	www.pami.ca
Corn	CORN	www.uaex.uada.edu
Edible Bean	ED BEAN	www.ag.ndsu.edu
Flax	FLAX	www.pami.ca
Lentil	LENTIL	www.pami.ca
Red Lentil	LENTIL(R)	www.pami.ca
Oats	OATS	www.pami.ca
Peanut Kernel	Peanut(K)	Modified Henderson equation
Peanut Pod	Peanut(P)	Modified Henderson equation
Peas	PEAS	www.pami.ca
Long Grain Rice	LG RICE	www.uaex.uada.edu
Sorghum	SORGHUM	www.uaex.uada.edu
Soybean	SOYBEAN	www.uaex.uada.edu
Confectionery Sunflower	SUNFLWR(C)	www.ag.ndsu.edu
Sunflower Hull	SUNFLWR(H)	www.pami.ca
Sunflower Kernel	SUNFLWR(K)	www.pami.ca
Sunflower (Oil Hybrids)	SUNFLWR(OIL)	www.ag.ndsu.edu
Sunflower Seed	SUNFLWR(S)	www.pami.ca
Durum Wheat	WHEAT(D)	www.uaex.uada.edu
Wheat (Hard)	WHEAT(H)	www.ag.ndsu.edu
Wheat (Soft)	WHEAT(S)	www.ag.ndsu.edu
Wheat (Hard Red Spring)	WHEAT(HRS)	www.pami.ca
Wheat (Hard Red Winter)	WHEAT(HRW)	www.pami.ca

LEGAL STUFF

Legal Stuff

Here is the legal stuff you should know about this project...

Copyrights, Licenses, and Trademarks

This project is the result of years of work to create, gather, and test both the concepts and the components. The hardware parts are basic, “off the shelf” parts that anyone can buy. The firmware code is a culmination of our (The Computer Connectors LLC) ideas and various hardware vendor computer code. In the following paragraphs, we will attempt to present you with an authentic representation of the work accomplished by and the legal requirements imposed by those vendors.

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Arduino



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AM2302 Sensor

[The AM2305 sensor uses the same code as the AM2302]

Avoid the temptation to use the cheaper AM2302. Yes, they are cheaper, but they are also less reliable and a lot more work to implement.

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The FCC

You have probably noticed all the small print that comes with the products you buy. You find everything from safety information to EMF noise/interference compliance statements. We mention those two topics because we need to make some statements about them here.

FCC – Federal Communications Commission

<https://www.fcc.gov/>

From their website:

“The Federal Communications Commission regulates interstate and international communications by radio, television, wire, satellite and cable in all 50 states, the District of Columbia and U.S. territories. An independent U.S. government agency overseen by Congress, the commission is the United States' primary authority for communications law, regulation and technological innovation. In its work facing economic opportunities and challenges associated with rapidly evolving advances in global communications, the agency capitalizes on its competencies in:

- *Promoting competition, innovation and investment in broadband services and facilities*
- *Supporting the nation's economy by ensuring an appropriate competitive framework for the unfolding of the communications revolution*
- *Encouraging the highest and best use of spectrum domestically and internationally*
- *Revising media regulations so that new technologies flourish alongside diversity and localism*
- *Providing leadership in strengthening the defense of the nation's communications infrastructure”*

The first statement relates to our motivation in establishing this project/product.

- **Promoting competition:** You have seen the advertisements... A company offers a product to accomplish all of the objects as outlined on this document's title page. All while only spending \$\$\$\$,\$\$\$,\$\$ (basically all of your profit) per bin on their product. Now, don't get us wrong. We do believe in profits. However, we also believe in a competitive marketplace along with a “fair profit.” There are too many monopolistic companies in too many industries. Effectively none of those monopolies has your wellbeing in mind. Thus, at least in this small project, **we believe that you should have the knowledge and tools to build your own system.**
- **Encouraging the highest and best use of [the] spectrum:** You can't see it, but all around you are packets of radiation from everywhere (including outer space). In fact, if it weren't for our atmosphere's ability to filter out a lot of those cosmic rays from all corners of the universe, we... wouldn't/couldn't exist.

Regarding **man-made** radiation, every electrical source/product produces some sort of radiation. Even your own body produces electrical impulses to energize your heart and other muscles. For this project, we will be using two potential sources of radiation. The first device is the “Arduino” processor board [a.k.a. Computer]. You should consider it the brains of this device. The memory components on this board store the firmware code while the processor and other components follow the instructions provided by the code.

*Side Note: If you want to learn more about how computers work, have a look at our book: “**How Computers Work.**”*

<https://author.amazon.com/claim/join?query=ronald+lind>

The second potential radiator is the LCD screen. It doesn't produce any power of its own, but it **may** have the potential to be a “secondary” radiator.

Now, you don't need to worry about the “radiation.” You would get a higher dose sitting in front of your TV in the evening than you would from being near these two devices. Plus, the radiation frequency is “non-ionizing.” https://en.wikipedia.org/wiki/Non-ionizing_radiation So, other than raising your body temperature by an immeasurable fraction of a degree, there is no threat to your body.

Regardless, when you buy these components, it is important to buy them: A) from a reputable vendor and B) that they have an FCC certification. Normally, I wouldn't do any vendor recommendations, but we rely heavily on Adafruit (www.adafruit.com) and Arduino (www.arduino.cc) for this project. So, hint: buy from them. 😊

Terms of Use

Please see our website (www.thecomputerconnectors.com/client-resources.html) for the latest version of our “Terms of Use.” It governs our relationship.

Privacy Policy

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Suitability of Purpose

As mentioned above, you need to be the subject matter expert regarding your grain storage. Continue to use your accumulated knowledge and expertise. This device simply gives you additional information upon which to base your decisions. The information this device provides, may or may not be useful in your specific situation.

Warranty

This project uses components that are sourced from various vendors. Each component may or may not have its own, vendor-backed warranty. Please check each of those vendors for any warranty coverage.

Any specific component that we may provide, will come with a warranty as described on our website (www.thecomputerconnectors.com/client-resources.html).

Acknowledgements

The Computer Connectors LLC would like to thank all those people and organizations that contributed to the success of this project. Without you, this project and the resulting device would not exist.

Safety

Everyone knows, and no one knows better than a farmer, that farming is a DANGEROUS occupation. According to the CDC, “...the agriculture industry is consistently at the highest risk for occupational injuries and fatalities, with 453 fatalities in 2021, or an equivalent of 19.5 deaths per 100,000 workers.”

Learn more on the CDC website at: www.cdc.gov/niosh/newsroom/feature/national-farm-safety-and-health.html

In everything that you do:

- Slow down!
- Evaluate what you are about to do!
- Ask yourself, what risk will my actions expose me to?
- How can I minimize those risks?
- Do NOT think that “An injury cannot or will not happen to me.” It definitely CAN!

We would encourage you to learn more about your risks. The CDC link above is one resource. Here is another: www.uaex.uada.edu/publications/pdf/FSA-1062.pdf.

As the old saying goes: **Safety is Priority ONE!**

(Un)Common – HMT

Make Your Own Grain Bin Monitoring System

USER'S MANUAL

User's Manual

Powering the Device

NOTE: We STRONGLY advise you to use the USB port to power this device. The other power connections on the circuit board are NOT adequately protected.

When you purchase the device, it normally comes with a USB cable – ensure that it does. We recommend connecting the USB cable to the following power sources:

- Wall charger: The kind you use to charge your phone.



- A portable battery specifically for charging/powering mobile devices (ex. Your phone)



Technically, you could use any power source that produces the same kind of power you use to charge your phone. Just make sure that it produces a steady, reliable +5 volts and can deliver at least 1.5amps. A surge protector would also be a great idea.

Using Portable Batteries

These batteries (mentioned above) come in all kinds of shapes and sizes. The commonalities that you must look for are:

- They will have one or more USB charging ports
- They will have a mAH (milli-Amp-Hour) rating.
 - Think of this rating as “The amount of energy it can hold.”
- You will also need to look for portable batteries that don’t turn themselves off.
 - We found that a really nice 17,000mAH pack would turn itself off automatically because it didn’t recognize any significant power draw from our HMT device.

We have done some **very minimal** testing with a couple of these devices. Here is what we have learned:

- They are super convenient
 - Take them anywhere

- For us, a 17,000mAH battery pack powered our HMT device for about 24 hours in a warm environment. Your experience will differ, AND in a cold environment, you should expect the battery pack to provide about ¼ to ½ less operational time.

Device Longevity

Theoretically, the device (sensors and cabling excluded) should last indefinitely. On our own farm and during development, we have left our system out in subzero cold for extended periods of time without damage. The thing we have noticed: In really cold weather, the LCD and its touch functions seem “sluggish.”

[See “Sensor Longevity” below.]

FREE Version

You are invited and encouraged to build and use as many “FREE” versions of this device as you want. There is no limitation on the number of devices you may build. HOWEVER, this product is copyright and trademark protected. Thus, **YOU MAY NOT SELL THE DEVICES YOU BUILD!**

A primary difference between the FREE and PRO versions is the number of allowed sensors. The FREE version allows you two sensors. One [the “Weather Sensor”] for monitoring the outside air, and one sensor for monitoring the grain inside your bin.

PRO Version

The PRO version has a few additional capabilities.

- It provides you with the ability to add two more sensors to the system for a total of four.
 - One for monitoring the outside air’s temperature and humidity
 - Three for monitoring the temperature and humidity of the grain inside your bin.
- It has the ability to power four “**Outputs**”
 - A “**Fan On Recommendation**” LED that lights up when the device determines that it may be beneficial to run your bin fan.
 - A “**Fan Off Recommendation**” LED that lights up when the device determines that you should not need to run your bin fan.
 - Two “**ACCESSORY**” outputs that can provide control signals to external, third-party equipment. This ACCESSORY port generally follows the same fan operation recommendation pattern as the “Fan On” LED mentioned above. The difference is simply that this output has a built-in OFF → ON delay of approximately 20 minutes to prevent the “quick cycling” of any external, third-party equipment.
- Grain industry standard moisture values are programmed into the PRO version’s recommendation LEDs and ACCESSORY port. Thus, the PRO version will not light up the green

LED (telling you to run the fan) if the device determines that the weather sensor's calculated equilibrium will take your grains moisture content further from the industry standard.

“PRO” License Number

When building and operating the PRO version, you will see references to a “License Number.” We will provide you with one or more add-on components that are necessary to convert your FREE version to a PRO version. Those add-on components will come with a unique “License Number” that you will use to pair the component(s) with the device you built.

With the FREE version, the “License Number” will usually display as “0.” However, not all MEGA devices are identical, and some may display numbers other than 0. This is not something to be concerned with. In the FREE version, the license number is irrelevant to its operation.

Sensors

Sensor Longevity

Unlike the HMT device itself, the sensors will have a limited life. After all, like you, they must do their job in a cold/hot, wet/dry, dusty/dirty environment. Even the manufacturer(s) tell you in the documentation that they will “drift” (become less accurate) over time. There is no one set timeline for replacing the sensors, but **we would strongly suggest replacing the sensors prior to grain fill EVERY YEAR.** After all, **they really are very cheap compared to the value of the job they do!**

An additional caution: The humidity portion of these sensors is sensitive to water immersion. Do not immerse your sensors in water.

Calibrating the System's Sensors

The other thing that the manufacturers tell us in the documentation is that no two sensors will read identically. Example, one may read 65 degrees while another may read 64 or 66.

That is why we designed the system to accommodate these differences by allowing you to set “calibration biases” for each sensor.

We will discuss the “Calibration” process a little later.

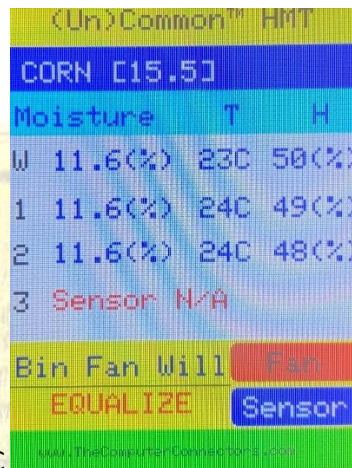
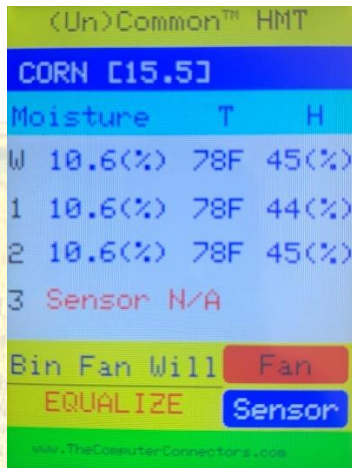
Using the System

NOTE: The system is designed to accommodate 24 different grain types. In future releases, we hope to add support for even more grain types. The current challenge is finding verifiable Equilibrium Moisture charts for those additional grains. If you grow crops other than those supported by this device, let us know where we can find and verify Temperature/Humidity/Moisture equilibrium values for your grain type.

IMPORTANT NOTE: Much of the work done to create the equilibrium moisture values presented by this device was completed in the 80's and 90's. Since that time, plant hybridization has dramatically changed the characteristics for nearly all commonly grown crops. Thus, the “Moisture” values that this device provides should only be used as a GUIDELINE for your grain conditioning efforts. In fact, we compared charts from two popular, commercially sold bin management systems [competitors to this device] and found that the grain’s moisture value readings were typically about 1% (ex. 13% vs. 14%) different (sometimes even more). Bottom line: **treat the Moisture readings only as reasonable ESTIMATES.**

ALSO: The moisture readings are highly dependent upon the accuracy of the sensors! We designed this system to allow for easy sensor replacement. Replace your sensors OFTEN!

The “Dashboard” Screen



Temp in either F or C

This is where you will find:

- The Moisture, Temperature, and Humidity values for each installed sensor
- The “Secret” spot where you can touch to change the grain type
 - Example: Barley → ... → Corn → ... → Soybeans → ... → (back to Barley)
- The “Secret” spot where you can switch between Fahrenheit and Celsius temperature values
- The “Secret” spot where you view or add the PRO license codes
- The “Fan Effect”
- The “Fan” setup” button
- The “Sensor” setup button

Let’s first start our “Dashboard” discussion with the “Secret Spots.”

The “Grain Type” Secret Spot

In the Dashboard image above, you will notice the “Corn” reference.

If you touch the word “Corn,” the system will switch to the next grain – Edible Bean.

As you continue to touch the grain type spot, the device cycles through the list of grains until it reaches the last type and cycles back to the beginning of the list – Barley.

The “Fahrenheit vs. Celsius” Secret Spot

In the image above, simply touch anywhere in the sensor display area to switch the display between Fahrenheit and Celsius.

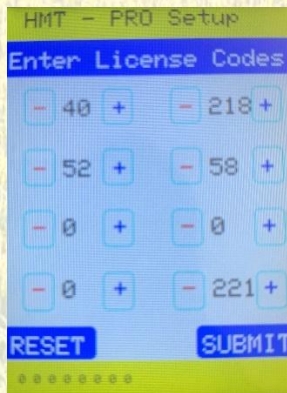
The “Licensing” Secret Spot

In the image above, notice the “HMT – PRO Edition” header.

If you haven’t already upgraded to PRO, you will see an “HMT – FREE Edition” header.

After you order and receive (from us) the PRO version upgrade component(s) and instructions, simply touch that header to access the Licensing Screen.

When you order the PRO version license, we will send you the necessary component(s), specific instructions on how to install the component(s), and the license numbers to configure in the License Screen (below).



Sensor Data

On the Dashboard image, notice the W, 1, 2, and 3 rows. Each of those rows represents the data read and calculated from an individual sensor.

The Grain Type and Standard Moisture

In the Dashboard image above, notice that there is a “[15.5]” after the word “CORN.” This is telling you that the industry standard moisture for corn is 15.5%. If you touch the grain type area several time until you reach “SOYBEAN” as your grain type, the device will tell you that the industry standard moisture for soybeans is [13] 13%.

This industry standard is programmed into the PRO version’s recommendation LEDs and ACCESSORY port. Thus, the PRO version will not light up the green LED (telling you to run the fan) if the device determines that the weather sensor’s calculated equilibrium will take your grains moisture content further from the industry standard.

IMPORTANT NOTE: These industry standard moisture values may not be “hard and fast” rules in your area/operation. If you regularly market your grain based upon different criteria, the “Recommendation LEDs” and “Accessory Ports” may not be relevant to your individual scenario.

Weather Sensor

In the Dashboard image, sensor “W” (the Weather sensor) is reading 78° Fahrenheit [25.6° Celsius], and 44-45% humidity. The device is using that data to calculate an equilibrium moisture content of 10.6% for Corn.

Equilibrium Moisture

Remember our previous discussion on “Equilibrium Moisture”?

Simply put, the Equilibrium Moisture calculation, is the moisture content your grain will “equalize” to when subjected to the outside air.

Let’s run through an **example**.

- When you put your corn in the bin at harvest, it was testing 18% moisture.
- You plan to use your bin’s fan to draw air through the corn to dry it down to the 15.5% industry standard.
 - [Chances are that you’ve been doing this for a lot of years].

But, how do you know that running the fan will help?

And, how do you know when to stop running the fan?

The HMT device you are building will help you answer those two questions.

If the outside air Weather sensor tells you that the current air condition is 78°F and 44% humidity, that means that the bin fan will eventually dry your corn to 10.6% moisture. Obviously, that is WAY below the industry standard of 15.5% for CORN. However, if you run the fan for a limited amount of time under these conditions, it will have the benefit of drying your 18% corn until you turn it off when the #1 - #3 sensors tell you that your corn has reached the industry standard 15.5% moisture.

Additionally, the PRO version will light up the green LED to tell you when it recommends running the bin fan. Similarly, it will light up the red LED to tell you when it recommends turning off the bin fan.

It should probably go without saying that the outside air conditions will change throughout the day and night. Thus, keep an eye out for these weather changes and adjust your fan operation decisions accordingly. For instance, if you are trying to dry your grain, you probably won’t be running your fan at night and sucking in all that dew to be absorbed by your grain.

Grain Sensors

In the Dashboard image, sensors 1 – 3’s purpose is to measure the Temperature and Humidity in the grain. (NOTE: In the image, sensor #3 is not connected.) The device then calculates the “Equilibrium Moisture” for the selected grain type (in this case – Corn).

(Un)Common™ HMT			
CORN [15.5]			
Moisture	T	H	
W 10.6(%)	78F	45(%)	
1 10.6(%)	78F	44(%)	
2 10.6(%)	78F	45(%)	
3	Sensor N/A		
Bin Fan Will		Fan	
EQUALIZE		Sensor	
www.TheComputerConnectors.com			

The Fan Effect

Just to the left of the Setup button, you will notice a message that will help you understand what effect the grain bin fan will have on your grain. There are four possible messages here:

1. DRY
2. REHYDRATE
3. EQUALIZE
4. N/A

DRY

This message is telling you that the outside air will DRY your grain if the bin fan is left to run. This “DRY” message appears if:

- The average of three grain sensors' (1 – 3) moisture content values is greater than 1 point higher than the Weather sensor's calculated moisture.

REHYDRATE

This message is telling you that the outside air will REHYDRATE (add moisture to) your grain if the bin fan is left to run. This “REHYDRATE” message appears if:

- The average of three grain sensors' (1 – 3) moisture content values is less than 1 point lower than the Weather sensor's calculated moisture.

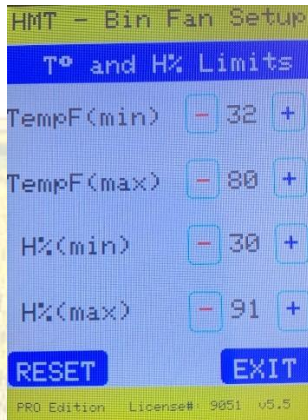
EQUALIZE

You have probably learned over the years that your grain's condition can vary from location to location within your bin. This message is telling you that the outside air will work to EQUALIZE the temperature and moisture of your grain if the bin fan is left to run.

NOTE: Recall how we previously explained that the moisture calculations provided by this (and all grain bin monitoring systems) should be considered an ESTIMATE of the grain's actual moisture. This “EQUALIZE” message appears if:

- The average of three grain sensors (1 – 3) moisture values is within 1 point (higher or lower) of the Weather sensor's calculated moisture.
 - Recall that the FREE version only has one grain sensor.

The “Fan Setup” Screen



From the “Dashboard” screen, a touch on the “Fan” button will take you to the setup screen for the Fan’s Min/Max Limits for Temperature(always in F) and Humidity. This is where you set the limits for:

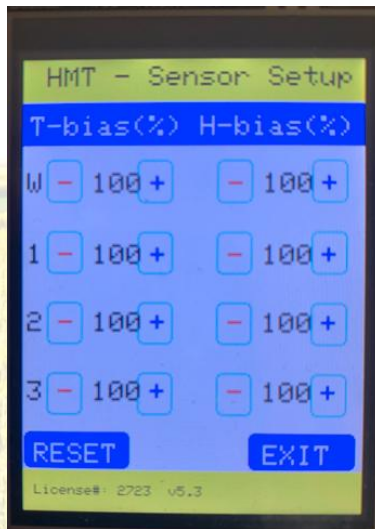
- The MINIMUM Temperature at which you want the HMT device to recommend running the bin fan.
- The MAXIMUM Temperature at which you want the HMT device to recommend running the bin fan.
- The MINIMUM Humidity at which you want the HMT device to recommend running the bin fan.
- The MAXIMUM Humidity at which you want the HMT device to recommend running the bin fan.

Just like the calibration settings (discussed below), it is important that you verify these numbers prior to each harvest season and then regularly adjust them throughout the storage season.

The “Sensor Setup” Screen

The Setup Button

From the “Dashboard” screen, a touch on the “Sensor” button will take you to the Sensor Setup screen.



Just as on the Dashboard screen, we have 4 rows to notice here. Each of those rows represent the “bias” setting for the respective sensor.

Allow us to provide an example...

Notice in row W that our “H” (humidity) bias is set to 100%. If that was set, for example, to 92%, that simply would mean that we had previously determined that our Weather sensor normally reads the humidity a few percentage points above reality. Thus, we would set the bias down to 92% to accommodate that inaccuracy. The HMT device then uses that adjusted sensor humidity value to make its moisture computations.

Similarly, if we had previously determined that our sensor 3 reads low on our grain’s temperature, we would bump up the “T” (temperature) bias setting on sensor 3 to accommodate that inaccuracy.

Sensor Calibration

Before you install the sensors in your bin, you will need to “calibrate” your sensors.

We also recommend testing and setting your sensors bias settings as often as you can – yes, even after installation.

Examples: If a sensor normally reads high on temperature, you need to reduce its bias (< 100%) setting. If a sensor reads low on humidity, you need to increase its bias (> 100%) setting.

To accomplish this:

1. Find yourself a reliable temperature and humidity sensing device.
 - a. **Suggestion: Build yourself a separate FREE unit that will always contain a new sensor and to which you can compare the previously installed production sensor values.**
2. One by one, bring this separate reference device and its sensor near each of the production sensors.

3. Pause for several minutes for the reference device to acclimate to the environment near the production sensor.
4. Now read the Temperature and Humidity from both the production and reference sensors
5. In the Setup screen of the production HMT device, adjust the “Bias” for the installed, production sensor up or down as necessary so that the adjusted Temperature and Humidity match that of the reference sensor.
6. Repeat the process for each installed, production sensor.

When you are done adjusting the sensor bias settings, simply press the “EXIT” button. The device will resume its normal operation (with the newly calibrated sensor data).

Appendix A: Troubleshooting

SYMPTOM	Possible Cause	Possible Solution	Additional Note
No sign of life	USB power supply failure	Try a different USB power supply	Some older, cheap cell phone chargers simply cannot supply enough amperage. Try a newer power supply.
		Try a different USB cable	
The LED on the MEGA board lights up but the LCD does not	Programming failed	Try to reload the device firmware	If the device was not successfully programmed, the LCD has nothing to display.
	Bent or broken pin connecting either the terminal board or the LCD to the MEGA board	During and after the build process, ensure that there are no bent or broken pins	
	The LCD was incorrectly layered on top of the terminal board	Ensure that the SD card slot on the bottom of the LCD is located near the USB connection on the MEGA board	
	The LCD is broken	Try replacing the LCD	After all, they are pretty cheap
	The MEGA board is broken	Try replacing the MEGA board	When operating correctly, the LED on the MEGA board will turn ON→OFF→ON every 30-60 seconds. (During the sensor read cycle.)

SYMPTOM	Possible Cause	Possible Solution	Additional Note
The LCD works but is very dim	Power issue	Try a different USB power supply	Some older, cheap cell phone chargers simply cannot supply enough amperage. Try a newer power supply that can deliver >1.5mA.
	[On PRO version] The "ACCESSORY" and LED ports are drawing too much power	Determine which component is drawing too much power.	One at a time, try disconnecting the LEDs and ACCESSORY connections
		Find a way to draw less power from the ACCESSORY ports.	Ex. Use a different external control circuit.
		Add larger resistors to the LED connections	This may decrease the brightness of the LEDs but it could also brighten the LCD.
The touch functions on the SETUP and LICENSING screens do not work	Bent or broken pin connecting either the terminal board or the LCD to the MEGA board	During and after the build process, ensure that there are no bent or broken pins	
	The LCD is broken or cracked	Try replacing the LCD	Even a small crack in the screen can impact the touch features.
The text and images on the screen do not match the touch locations.	Recently, we received notification from a few builders that their TFT screens were incorrectly displaying all text and images. In these cases, everything was working except that all the text and images were rotated 180° from the touch locations.	We added an "orientation jumper" feature to the software. Now, you can jumper either pins 33-34 or 34-35 to change the orientation so that the text and images match your touch locations.	

SYMPTOM	Possible Cause	Possible Solution	Additional Note
Sensor Read Failure	<i>MOST COMMON:</i> Wiring issue	Verify all of the sensor wire connections	
		Sensor not receiving +5V power or proper Ground	Measure the +5V and Ground wire connections at the sensor.
		Sensor data wire not properly connected to terminal block	Use Ohm meter to ensure that the data wire between the sensor and the terminal block has continuity.
		The sensor cable is too long	We have tested sensors with cables up to 100'. However, we do not recommend cables over 50 – 70 feet. When the cable gets too long and/or is routed near electrical power lines, the data signal becomes inconsistent.
	Bent or broken pin connecting either the terminal board to the MEGA board	During and after the build process, ensure that there are no bent or broken pins	
	The sensor is broken	Try replacing the sensor	
The sensor is reading incorrectly	The sensor bias settings need to be updated	See the “Calibration” section in this document	
	The sensor is broken	Try replacing the sensor	
Sensor says “Out of Range”	The sensor is telling you that the environmental conditions are outside of the device’s Temperature and/or Humidity range.	Wait for better weather.	

SYMPTOM	Possible Cause	Possible Solution	Additional Note
The Green LED is lit	This means that the HMT device believes that your grain could benefit from running your bin's fan.	Consider your experience and common sense... Decide whether you want to run the fan.	
The Red LED is lit	This means that the HMT device believes that there would be no benefit from running your bin's fan.	Consider your experience and common sense... Decide whether you want to run the fan.	
	The weather sensor's MIN and MAX settings for temperature and humidity are outside of your set ranges.	Change the MIN and MAX temperature and humidity in the fan setup screen.	
	"Out of Range" displayed on screen because the air temperature and/or humidity are out of range.	Adjust the range of allowed temperature and humidity values in the fan setup screen.	
Both the Green and the Red LEDs are lit up	This means that the HMT device believes that your grain could benefit from running your bin's fan. HOWEVER, if you are using one of the HMT device's ACCESSORY ports, there is an approximately 20 minute delay between when the device lights up the Green LED and when it energizes the ACCESSORY port.	Consider your experience and common sense... Decide whether you want to run the fan.	

SYMPTOM	Possible Cause	Possible Solution	Additional Note
None of the LEDs are lit	The LEDs only operate in the PRO version.	Upgrade to the PRO version if you haven't already.	
	One or more of the LEDs is broken.	Use a volt meter to check the voltage on pins 38, 40, and 42. If you receive > +3.5v on either of those pins, replace the LED connected to that pin.	
	You don't receive > +3.5v on any of the LED pins.	Try a different USB power supply	
		During and after the build process, ensure that there are no bent or broken pins	
The "Bin Fan Will" area is displaying "*AirT<minT*" or "*AirT>maxT*"	This is telling you that the environmental air Temperature is above or below the minimum or maximum temperature at which the fan is allowed to run.	Option 1: Verify that the minimum and maximum Fan temperature settings are correct. Option 2: Wait for better weather.	
The "Bin Fan Will" area is displaying "*AirH<minH*" or "*AirH>maxH*"	This is telling you that the environmental air Humidity is above or below the minimum or maximum humidity at which the fan is allowed to run.	Option 1: Verify that the minimum and maximum Fan Humidity settings are correct. Option 2: Wait for better weather.	

Barley

BARLEY																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	NULL	11.8	12.6	13.3	14.0	14.9	15.6	16.6	17.5	18.6	19.8	21.2	NULL	-1.1
35	NULL	NULL	NULL	NULL	NULL	NULL	10.7	11.4	12.1	12.8	13.6	14.4	15.1	16.0	17.0	18.2	19.6	NULL	NULL	1.7
40	NULL	NULL	NULL	NULL	NULL	NULL	10.4	11.1	11.8	12.5	13.3	14.0	14.8	15.7	16.7	17.9	19.2	NULL	NULL	4.4
45	NULL	NULL	NULL	NULL	NULL	NULL	10.2	10.9	11.5	12.2	13.0	13.8	14.5	15.4	16.4	17.6	18.9	NULL	NULL	7.2
50	NULL	NULL	NULL	NULL	NULL	NULL	10.3	11.1	11.8	12.5	13.3	14.1	14.9	15.8	16.8	18.0	19.4	NULL	NULL	10.0
55	NULL	NULL	NULL	NULL	NULL	NULL	10.0	10.8	11.5	12.2	13.0	13.8	14.6	15.5	16.6	17.7	19.1	NULL	NULL	12.8
60	NULL	NULL	NULL	NULL	NULL	NULL	10.2	10.9	11.7	12.4	13.2	14.0	14.8	15.8	16.8	18.0	19.4	NULL	NULL	15.6
65	NULL	NULL	NULL	NULL	NULL	NULL	9.9	10.7	11.4	12.1	12.9	13.7	14.5	15.5	16.6	17.8	19.2	NULL	NULL	18.3
70	NULL	NULL	NULL	NULL	NULL	NULL	9.4	10.1	10.9	11.5	12.3	13.1	13.9	14.9	15.8	17.0	18.4	NULL	NULL	21.1
75	NULL	NULL	NULL	NULL	NULL	NULL	8.9	9.6	10.3	11.0	11.8	12.6	13.3	14.3	15.2	16.4	17.7	NULL	NULL	23.9
80	NULL	NULL	NULL	NULL	NULL	NULL	8.9	9.7	10.4	11.2	11.9	12.7	13.5	14.5	15.4	16.6	18.0	NULL	NULL	26.7
85	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	29.4
90	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	32.2
95	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	35.0
100	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	37.8
Adapted from Source: https://pami.ca/wp-content/uploads/2024/01/NAD2BFactsheet_FINAL_2020-08-24-1.pdf																				
Downloaded 2024May3																				
Equation: Modified Chung-Pfost																				
(NULL: Out of Chart Range from Original Data Source)																				

Canola

CANOLA																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	7.1	7.9	8.7	9.4	10.2	11.1	11.9	12.9	13.9	15.1	16.6	NULL	NULL	-1.1
35	NULL	NULL	NULL	NULL	NULL	NULL	6.3	6.9	7.6	8.3	8.9	9.7	10.5	11.4	12.3	13.4	14.6	NULL	NULL	1.7
40	NULL	NULL	NULL	NULL	NULL	NULL	6.0	6.6	7.2	7.9	8.6	9.3	10.0	10.8	11.7	12.8	14.0	NULL	NULL	4.4
45	NULL	NULL	NULL	NULL	NULL	NULL	5.7	6.3	6.9	7.6	8.2	8.9	9.6	10.4	11.3	12.2	13.4	NULL	NULL	7.2
50	NULL	NULL	NULL	NULL	NULL	NULL	5.7	6.3	7.0	7.6	8.3	8.9	9.7	10.5	11.3	12.3	13.5	NULL	NULL	10.0
55	NULL	NULL	NULL	NULL	NULL	NULL	5.5	6.1	6.7	7.2	7.9	8.5	9.3	10.0	10.9	11.8	13.0	NULL	NULL	12.8
60	NULL	NULL	NULL	NULL	NULL	NULL	5.5	6.1	6.7	7.3	7.9	8.6	9.4	10.1	10.9	11.9	13.0	NULL	NULL	15.6
65	NULL	NULL	NULL	NULL	NULL	NULL	5.2	5.9	6.5	7.1	7.7	8.3	9.0	9.7	10.5	11.4	12.5	NULL	NULL	18.3
70	NULL	NULL	NULL	NULL	NULL	NULL	4.9	5.5	6.0	6.6	7.1	7.7	8.3	9.1	9.8	10.7	11.7	NULL	NULL	21.1
75	NULL	NULL	NULL	NULL	NULL	NULL	4.6	5.1	5.6	6.2	6.7	7.2	7.8	8.5	9.2	10.0	11.0	NULL	NULL	23.9
80	NULL	NULL	NULL	NULL	NULL	NULL	4.7	5.1	5.6	6.1	6.7	7.3	7.9	8.5	9.2	10.1	11.1	NULL	NULL	26.7
85	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	29.4
90	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	32.2
95	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	35.0
100	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	37.8

Adapted from Source: https://pami.ca/wp-content/uploads/2024/01/NAD2BFactsheet_FINAL_2020-08-24-1.pdf

Equation: Modified Henderson

Downloaded 2024May3

(NULL: Out of Chart Range from Original Data Source)

Corn

CORN																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-1.1
35	NULL	NULL	NULL	NULL	9.3	10.3	11.2	12.1	13.0	13.9	14.8	15.7	16.6	17.6	18.7	19.8	21.2	22.9	NULL	1.7
40	NULL	NULL	NULL	NULL	9.1	10.0	10.9	11.8	12.7	13.5	14.4	15.3	16.2	17.1	18.2	19.3	20.7	22.3	NULL	4.4
45	NULL	NULL	NULL	NULL	8.8	9.8	10.6	11.5	12.3	13.2	14.0	14.9	15.8	16.7	17.7	18.9	20.2	21.8	NULL	7.2
50	NULL	NULL	NULL	NULL	8.6	9.5	10.4	11.2	12.0	12.9	13.7	14.5	15.4	16.3	17.3	18.5	19.8	21.4	NULL	10.0
55	NULL	NULL	NULL	NULL	8.4	9.3	10.1	11.0	11.8	12.6	13.4	14.2	15.1	16.0	17.0	18.1	19.3	20.9	NULL	12.8
60	NULL	NULL	NULL	NULL	8.2	9.1	9.9	10.7	11.5	12.3	13.1	13.9	14.8	15.7	16.6	17.7	18.9	20.5	NULL	15.6
65	NULL	NULL	NULL	NULL	8.0	8.9	9.7	10.5	11.3	12.0	12.8	13.6	14.5	15.3	16.3	17.4	18.6	20.1	NULL	18.3
70	NULL	NULL	NULL	NULL	7.9	8.7	9.5	10.3	11.0	11.8	12.6	13.4	14.2	15.0	16.0	17.0	18.2	19.8	NULL	21.1
75	NULL	NULL	NULL	NULL	7.7	8.5	9.3	10.1	10.8	11.6	12.3	13.1	13.9	14.8	15.7	16.7	17.9	19.4	NULL	23.9
80	NULL	NULL	NULL	NULL	7.6	8.4	9.1	9.9	10.6	11.4	12.1	12.9	13.7	14.5	15.4	16.4	17.6	19.1	NULL	26.7
85	NULL	NULL	NULL	NULL	7.4	8.2	9.0	9.7	10.4	11.2	11.9	12.6	13.4	14.3	15.2	16.2	17.3	18.8	NULL	29.4
90	NULL	NULL	NULL	NULL	7.3	8.1	8.8	9.5	10.3	11.0	11.7	12.4	13.2	14.0	14.9	15.9	17.0	18.5	NULL	32.2
95	NULL	NULL	NULL	NULL	7.2	7.9	8.7	9.4	10.1	10.8	11.5	12.2	13.0	13.8	14.7	15.6	16.8	18.2	NULL	35.0
100	NULL	NULL	NULL	NULL	7.1	7.8	8.5	9.2	9.9	10.6	11.3	12.0	12.8	13.6	14.5	15.4	16.5	17.9	NULL	37.8
Data Source: https://www.uaex.uada.edu/farm-ranch/crops-commercial-horticulture/Grain_drying_and_storage/Docs/FSA 1074 Grain Drying Tools Equilibrium Moisture Content Tables and Psychrometric Charts.pdf																				
Downloaded 2024May3																				
Equation: (Unkown)																				
(NULL: Out of Chart Range from Original Data Source)																				

Edible Bean

EDIBLE BEAN																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	3.30	4.70	5.90	6.90	7.80	8.70	9.50	10.30	11.10	11.90	12.70	13.50	14.40	15.30	16.30	17.50	18.70	20.40	22.70	-6.7
25	3.30	4.70	5.80	6.80	7.80	8.60	9.40	10.20	11.00	11.80	12.60	13.50	14.30	15.30	16.30	17.40	18.70	20.20	22.60	-3.9
30	3.20	4.70	5.80	6.80	7.70	8.60	9.40	10.20	11.00	11.80	12.60	13.40	14.30	15.20	16.20	17.30	18.60	20.20	22.50	-1.1
35	3.20	4.60	5.80	6.80	7.70	8.50	9.30	10.10	10.90	11.70	12.50	13.30	14.20	15.10	16.10	17.20	18.50	20.10	22.40	1.7
40	3.20	4.60	5.80	6.70	7.60	8.50	9.30	10.10	10.90	11.70	12.50	13.30	14.10	15.00	16.00	17.10	18.40	20.00	22.30	4.4
45	3.20	4.60	5.70	6.70	7.60	8.40	9.20	10.00	10.80	11.60	12.40	13.20	14.10	15.00	15.90	17.00	18.30	19.90	22.20	7.2
50	3.20	4.60	5.70	6.70	7.60	8.40	9.20	10.00	10.80	11.50	12.30	13.10	14.00	14.90	15.90	17.00	18.20	19.80	22.10	10.0
55	3.10	4.60	5.70	6.60	7.50	8.40	9.20	9.90	10.70	11.50	12.30	13.10	13.90	14.80	15.80	16.90	18.10	19.70	22.00	12.8
60	3.10	4.50	5.60	6.60	7.50	8.30	9.10	9.90	10.70	11.40	12.20	13.00	13.90	14.80	15.70	16.80	18.10	19.60	21.90	15.6
65	3.10	4.50	5.60	6.60	7.40	8.30	9.10	9.80	10.60	11.40	12.20	13.00	13.80	14.70	15.70	16.70	18.00	19.50	21.80	18.3
70	3.10	4.50	5.60	6.50	7.40	8.20	9.00	9.80	10.60	11.30	12.10	12.90	13.70	14.60	15.60	16.70	17.90	19.50	21.80	21.1
75	3.10	4.50	5.50	6.50	7.40	8.20	9.00	9.70	10.50	11.30	12.00	12.80	13.70	14.60	15.50	16.60	17.80	19.40	21.70	23.9
80	3.10	4.40	5.50	6.50	7.30	8.20	8.90	9.70	10.50	11.20	12.00	12.80	13.60	14.50	15.40	16.50	17.70	19.30	21.60	26.7
85	3.10	4.40	5.50	6.40	7.30	8.10	8.90	9.70	10.40	11.20	11.90	12.70	13.50	14.40	15.40	16.40	17.70	19.20	21.50	29.4
90	3.00	4.40	5.50	6.40	7.30	8.10	8.80	9.60	10.40	11.10	11.90	12.70	13.50	14.40	15.30	16.40	17.60	19.10	21.40	32.2
95	3.00	4.40	5.40	6.40	7.20	8.00	8.80	9.60	10.30	11.10	11.80	12.60	13.40	14.30	15.20	16.30	17.50	19.10	21.30	35.0
100	3.00	4.30	5.40	6.30	7.20	8.00	8.80	9.50	10.30	11.00	11.80	12.60	13.40	14.20	15.20	16.20	17.40	19.00	21.20	37.8
Download Source : https://www.ag.ndsu.edu/graindrying/documents/GrainEquilibriumMoistureContentCharts.pdf																				
Equation: ASAE 245.4 Modified Henderson																				
Downloaded 2024May3																				
Dr. Kenneth J Hellevang, PE																				
NDSU Extension Service																				
Fargo, ND 58105																				

Flax

FLAX																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	NULL	7.8	8.5	9.0	9.5	10.0	10.6	11.2	11.8	12.6	13.4	14.4	NULL	-1.1
35	NULL	NULL	NULL	NULL	NULL	NULL	NULL	7.0	7.5	8.0	8.5	8.9	9.5	10.0	10.6	11.3	12.1	13.0	NULL	1.7
40	NULL	NULL	NULL	NULL	NULL	NULL	NULL	6.7	7.2	7.7	8.2	8.7	9.2	9.8	10.3	10.9	11.7	12.6	NULL	4.4
45	NULL	NULL	NULL	NULL	NULL	NULL	NULL	6.5	7.0	7.5	8.0	8.4	8.9	9.4	10.0	10.7	11.4	12.3	NULL	7.2
50	NULL	NULL	NULL	NULL	NULL	NULL	NULL	6.6	7.1	7.5	8.0	8.5	9.0	9.6	10.2	10.8	11.6	12.5	NULL	10.0
55	NULL	NULL	NULL	NULL	NULL	NULL	NULL	6.4	6.9	7.2	7.7	8.2	8.7	9.3	9.9	10.5	11.3	12.2	NULL	12.8
60	NULL	NULL	NULL	NULL	NULL	NULL	NULL	6.4	6.8	7.3	7.8	8.3	8.8	9.5	10.0	10.7	11.4	12.3	NULL	15.6
65	NULL	NULL	NULL	NULL	NULL	NULL	NULL	6.2	6.7	7.1	7.6	8.1	8.6	9.2	9.7	10.4	11.1	12.0	NULL	18.3
70	NULL	NULL	NULL	NULL	NULL	NULL	NULL	5.8	6.3	6.6	7.1	7.6	8.1	8.6	9.2	9.8	10.6	11.4	NULL	21.1
75	NULL	NULL	NULL	NULL	NULL	NULL	NULL	5.4	5.9	6.3	6.8	7.2	7.7	8.2	8.8	9.3	10.0	10.9	NULL	23.9
80	NULL	NULL	NULL	NULL	NULL	NULL	NULL	5.4	5.9	6.3	6.8	7.3	7.8	8.3	8.8	9.4	10.1	11.0	NULL	26.7
85	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	29.4
90	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	32.2
95	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	35.0
100	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	37.8
Adapted from Source: https://pami.ca/wp-content/uploads/2024/01/NAD2BFactsheet_FINAL_2020-08-24-1.pdf																				
Downloaded 2024May3																				
Equation: Modified Henderson																				
(NULL: Out of Chart Range from Original Data Source)																				

Lentil

LENTILS																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	9.8	10.9	11.8	13.0	14.2	15.4	16.9	18.6	20.6	23.1	26.4	NULL	NULL	-1.1
35	NULL	NULL	NULL	NULL	NULL	NULL	8.9	9.8	10.8	11.8	12.9	14.1	15.4	17.0	18.8	21.1	24.1	NULL	NULL	1.7
40	NULL	NULL	NULL	NULL	NULL	NULL	8.8	9.7	10.5	11.5	12.6	13.8	15.0	16.6	18.3	20.6	23.6	NULL	NULL	4.4
45	NULL	NULL	NULL	NULL	NULL	NULL	8.7	9.6	10.5	11.5	12.5	13.7	15.0	16.5	18.1	20.4	23.4	NULL	NULL	7.2
50	NULL	NULL	NULL	NULL	NULL	NULL	8.5	9.5	10.4	11.3	12.4	13.5	14.8	16.3	18.0	20.3	23.3	NULL	NULL	10.0
55	NULL	NULL	NULL	NULL	NULL	NULL	8.5	9.4	10.3	11.3	12.3	13.5	14.7	16.2	17.9	20.2	23.2	NULL	NULL	12.8
60	NULL	NULL	NULL	NULL	NULL	NULL	8.4	9.3	10.2	11.2	12.2	13.4	14.5	16.0	17.8	20.1	23.0	NULL	NULL	15.6
65	NULL	NULL	NULL	NULL	NULL	NULL	8.4	9.2	10.1	11.0	12.1	13.2	14.4	15.9	17.7	19.9	22.8	NULL	NULL	18.3
70	NULL	NULL	NULL	NULL	NULL	NULL	7.9	8.7	9.5	10.5	11.4	12.5	13.7	15.1	16.7	18.9	21.6	NULL	NULL	21.1
75	NULL	NULL	NULL	NULL	NULL	NULL	7.5	8.3	9.1	10.0	10.9	11.9	13.1	14.5	16.2	18.1	20.8	NULL	NULL	23.9
80	NULL	NULL	NULL	NULL	NULL	NULL	7.4	8.2	9.0	9.9	10.9	11.8	12.9	14.3	16.0	17.9	20.6	NULL	NULL	26.7
85	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	12.9	14.3	16.0	17.9	20.6	NULL	NULL	29.4
90	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	32.2
95	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	35.0
100	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	37.8
Adapted from Source: https://pami.ca/wp-content/uploads/2024/01/NAD28Factsheet_FINAL_2020-08-24-1.pdf																				
Downloaded 2024May3																				
Equation: Modified Oswin																				
(NULL: Out of Chart Range from Original Data Source)																				

Lentil (Red)

LENTILS (RED)																					
Calculated Equilibrium Moisture																					
Temperature(F)	Humidity(%)																			Temperature(C)	
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%		
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7	
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9	
30	NULL	NULL	NULL	NULL	NULL	NULL	NULL	9.4	10.0	10.6	11.2	11.8	12.7	13.5	14.6	15.8	17.4	19.6	NULL	NULL	-1.1
35	NULL	NULL	NULL	NULL	NULL	NULL	NULL	8.7	9.1	9.6	10.2	10.8	11.6	12.4	13.3	14.5	15.9	18.0	NULL	NULL	1.7
40	NULL	NULL	NULL	NULL	NULL	NULL	NULL	8.5	9.0	9.5	10.1	10.6	11.4	12.1	13.1	14.1	15.7	17.7	NULL	NULL	4.4
45	NULL	NULL	NULL	NULL	NULL	NULL	NULL	8.4	8.9	9.4	10.0	10.5	11.3	12.0	12.9	13.9	15.6	17.6	NULL	NULL	7.2
50	NULL	NULL	NULL	NULL	NULL	NULL	NULL	8.4	8.8	9.4	9.9	10.5	11.3	11.9	12.9	13.9	15.5	17.5	NULL	NULL	10.0
55	NULL	NULL	NULL	NULL	NULL	NULL	NULL	8.3	8.7	9.3	9.8	10.4	11.2	11.9	12.8	13.8	15.4	17.4	NULL	NULL	12.8
60	NULL	NULL	NULL	NULL	NULL	NULL	NULL	8.3	8.7	9.2	9.8	10.4	11.1	11.8	12.8	13.6	15.3	17.2	NULL	NULL	15.6
65	NULL	NULL	NULL	NULL	NULL	NULL	NULL	8.2	8.6	9.1	9.7	10.3	11.0	11.8	12.7	13.3	15.2	17.1	NULL	NULL	18.3
70	NULL	NULL	NULL	NULL	NULL	NULL	NULL	7.8	8.3	8.8	9.3	9.9	10.5	11.2	12.1	13.2	14.6	16.4	NULL	NULL	21.1
75	NULL	NULL	NULL	NULL	NULL	NULL	NULL	7.6	8.0	8.4	8.9	9.5	10.2	10.9	11.7	12.7	14.3	15.9	NULL	NULL	23.9
80	NULL	NULL	NULL	NULL	NULL	NULL	NULL	7.5	7.9	8.4	8.9	9.5	10.1	10.8	11.7	12.7	14.1	15.7	NULL	NULL	26.7
85	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	29.4
90	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	32.2
95	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	35.0
100	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	37.8
Adapted from Source: https://pami.ca/wp-content/uploads/2024/01/NAD2BFactsheet_FINAL_2020-08-24-1.pdf																					
Downloaded 2024May3																					
Equation: Modified Halsey																					
(NULL: Out of Chart Range from Original Data Source)																					

Oats

OATS																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	10.8	11.3	11.8	12.5	13.0	13.6	14.3	14.9	15.6	16.6	17.6	NULL	NULL	-1.1
35	NULL	NULL	NULL	NULL	NULL	NULL	9.6	10.1	10.7	11.2	11.7	12.3	12.9	13.5	14.3	15.0	16.1	NULL	NULL	1.7
40	NULL	NULL	NULL	NULL	NULL	NULL	9.4	9.9	10.3	10.8	11.4	11.9	12.5	13.2	13.9	14.8	15.8	NULL	NULL	4.4
45	NULL	NULL	NULL	NULL	NULL	NULL	9.1	9.7	10.2	10.7	11.3	11.8	12.4	13.0	13.8	14.7	15.5	NULL	NULL	7.2
50	NULL	NULL	NULL	NULL	NULL	NULL	9.0	9.5	10.0	10.6	11.1	11.7	12.3	12.9	13.7	14.6	15.4	NULL	NULL	10.0
55	NULL	NULL	NULL	NULL	NULL	NULL	8.9	9.4	9.9	10.5	11.0	11.6	12.2	12.8	13.6	14.5	15.3	NULL	NULL	12.8
60	NULL	NULL	NULL	NULL	NULL	NULL	8.8	9.3	9.8	10.4	10.9	11.5	12.1	12.8	13.5	14.4	15.2	NULL	NULL	15.6
65	NULL	NULL	NULL	NULL	NULL	NULL	8.6	9.2	9.7	10.2	10.8	11.4	12.0	12.7	13.4	14.3	15.0	NULL	NULL	18.3
70	NULL	NULL	NULL	NULL	NULL	NULL	8.1	8.6	9.1	9.7	10.2	10.8	11.3	12.0	12.8	13.7	14.8	NULL	NULL	21.1
75	NULL	NULL	NULL	NULL	NULL	NULL	7.7	8.2	8.7	9.2	9.8	10.4	11.1	11.7	12.4	13.3	14.3	NULL	NULL	23.9
80	NULL	NULL	NULL	NULL	NULL	NULL	7.6	8.2	8.7	9.1	9.7	10.3	10.9	11.5	12.3	13.1	14.1	NULL	NULL	26.7
85	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	29.4
90	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	32.2
95	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	35.0
100	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	37.8

Adapted from Source: https://pami.ca/wp-content/uploads/2024/01/NAD2BFactsheet_FINAL_2020-08-24-1.pdf

Downloaded 2024May3

Equation: Modified Oswin

(NULL: Out of Chart Range from Original Data Source)

Peanut(Kernel)

PEANUT (Kernel)																			
Calculated Equilibrium Moisture																			
Temperature(F)	Humidity(%)																		Temperature(C)
	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	
15	2.5	3.3	4.1	4.9	5.6	6.4	7.2	8.0	8.8	9.6	10.6	11.6	12.7	13.9	15.4	17.2	19.6	23.3	-9.4
20	2.4	3.2	3.9	4.7	5.4	6.1	6.9	7.6	8.4	9.2	10.1	11.1	12.1	13.3	14.7	16.5	18.7	22.3	-6.7
25	2.3	3.1	3.8	4.5	5.2	5.9	6.6	7.3	8.1	8.9	9.7	10.6	11.7	12.8	14.2	15.8	18.0	21.4	-3.9
30	2.2	2.9	3.6	4.3	5.0	5.7	6.3	7.0	7.8	8.5	9.4	10.2	11.2	12.3	13.6	15.2	17.3	20.6	-1.1
35	2.1	2.8	3.5	4.2	4.8	5.4	6.1	6.8	7.5	8.2	9.0	9.9	10.8	11.9	13.1	14.7	16.7	19.9	1.7
40	2.1	2.7	3.4	4.0	4.6	5.3	5.9	6.5	7.2	7.9	8.7	9.5	10.4	11.5	12.7	14.2	16.1	19.2	4.4
45	2.0	2.7	3.3	3.9	4.5	5.1	5.7	6.3	7.0	7.7	8.4	9.2	10.1	11.1	12.3	13.7	15.6	18.6	7.2
50	1.9	2.6	3.2	3.8	4.4	4.9	5.5	6.1	6.8	7.5	8.2	8.9	9.8	10.8	11.9	13.3	15.1	18.0	10.0
55	1.9	2.5	3.1	3.7	4.2	4.8	5.4	6.0	6.6	7.2	7.9	8.7	9.5	10.4	11.5	12.9	14.7	17.5	12.8
60	1.8	2.4	3.0	3.6	4.1	4.7	5.2	5.8	6.4	7.0	7.7	8.4	9.2	10.2	11.2	12.5	14.2	17.0	15.6
65	1.8	2.4	2.9	3.5	4.0	4.5	5.1	5.6	6.2	6.8	7.5	8.2	9.0	9.9	10.9	12.2	13.9	16.5	18.3
70	1.7	2.3	2.8	3.4	3.9	4.4	4.9	5.5	6.1	6.7	7.3	8.0	8.8	9.6	10.6	11.9	13.5	16.1	21.1
75	1.7	2.2	2.8	3.3	3.8	4.3	4.8	5.4	5.9	6.5	7.1	7.8	8.5	9.4	10.4	11.6	13.2	15.7	23.9
80	1.6	2.2	2.7	3.2	3.7	4.2	4.7	5.2	5.8	6.3	6.9	7.6	8.3	9.2	10.1	11.3	12.8	15.3	26.7
85	1.6	2.1	2.6	3.1	3.6	4.1	4.6	5.1	5.6	6.2	6.8	7.4	8.1	8.9	9.9	11.0	12.5	15.0	29.4
90	1.6	2.1	2.6	3.1	3.5	4.0	4.5	5.0	5.5	6.0	6.6	7.3	8.0	8.7	9.7	10.8	12.3	14.6	32.2
95	1.5	2.0	2.5	3.0	3.5	3.9	4.4	4.9	5.4	5.9	6.5	7.1	7.8	8.5	9.4	10.5	12.0	14.3	35.0
100	1.5	2.0	2.5	2.9	3.4	3.8	4.3	4.8	5.3	5.8	6.3	7.0	7.6	8.4	9.2	10.3	11.7	14.0	37.8
Data Source Equation: Modified Henderson																			

Peanut(Pod)

PEANUT (Pod)																			
Calculated Equilibrium Moisture																			
Temperature(F)	Humidity(%)																		Temperature(C)
	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	
15	6.5	7.7	8.7	9.6	10.5	11.3	12.1	12.8	13.6	14.4	15.2	16.0	16.9	17.9	19.0	20.2	21.8	24.2	-9.4
20	6.0	7.1	8.1	9.0	9.7	10.5	11.2	11.9	12.7	13.4	14.1	14.9	15.7	16.6	17.6	18.8	20.3	22.5	-6.7
25	5.7	6.7	7.6	8.4	9.2	9.9	10.6	11.2	11.9	12.6	13.3	14.0	14.8	15.7	16.6	17.7	19.1	21.2	-3.9
30	5.4	6.4	7.2	8.0	8.7	9.4	10.0	10.7	11.3	11.9	12.6	13.3	14.1	14.9	15.8	16.8	18.1	20.1	-1.1
35	5.1	6.1	6.9	7.6	8.3	8.9	9.6	10.2	10.8	11.4	12.0	12.7	13.4	14.2	15.0	16.0	17.3	19.2	1.7
40	4.9	5.8	6.6	7.3	8.0	8.6	9.2	9.8	10.3	10.9	11.6	12.2	12.9	13.6	14.4	15.4	16.6	18.4	4.4
45	4.7	5.6	6.4	7.0	7.7	8.3	8.8	9.4	10.0	10.5	11.1	11.7	12.4	13.1	13.9	14.8	16.0	17.8	7.2
50	4.6	5.4	6.2	6.8	7.4	8.0	8.5	9.1	9.6	10.2	10.8	11.3	12.0	12.7	13.4	14.3	15.5	17.2	10.0
55	4.4	5.3	6.0	6.6	7.2	7.7	8.3	8.8	9.3	9.9	10.4	11.0	11.6	12.3	13.0	13.9	15.0	16.6	12.8
60	4.3	5.1	5.8	6.4	7.0	7.5	8.0	8.5	9.1	9.6	10.1	10.7	11.3	11.9	12.6	13.5	14.5	16.1	15.6
65	4.2	5.0	5.6	6.2	6.8	7.3	7.8	8.3	8.8	9.3	9.8	10.4	11.0	11.6	12.3	13.1	14.2	15.7	18.3
70	4.1	4.9	5.5	6.1	6.6	7.1	7.6	8.1	8.6	9.1	9.6	10.1	10.7	11.3	12.0	12.8	13.8	15.3	21.1
75	4.0	4.7	5.4	5.9	6.5	7.0	7.4	7.9	8.4	8.9	9.4	9.9	10.4	11.0	11.7	12.5	13.5	14.9	23.9
80	3.9	4.6	5.2	5.8	6.3	6.8	7.3	7.7	8.2	8.7	9.2	9.7	10.2	10.8	11.4	12.2	13.2	14.6	26.7
85	3.8	4.5	5.1	5.7	6.2	6.7	7.1	7.6	8.0	8.5	9.0	9.5	10.0	10.6	11.2	11.9	12.9	14.3	29.4
90	3.7	4.4	5.0	5.6	6.1	6.5	7.0	7.4	7.9	8.3	8.8	9.3	9.8	10.3	11.0	11.7	12.6	14.0	32.2
95	3.7	4.4	4.9	5.5	5.9	6.4	6.8	7.3	7.7	8.2	8.6	9.1	9.6	10.2	10.8	11.5	12.4	13.8	35.0
100	3.6	4.3	4.9	5.4	5.8	6.3	6.7	7.2	7.6	8.0	8.5	8.9	9.4	10.0	10.6	11.3	12.2	13.5	37.8
Data Source Equation: Modified Henderson																			

Peas

PEAS																					
Calculated Equilibrium Moisture																					
Temperature(F)	Humidity(%)																				Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%		
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	9.8	11.0	12.1	13.3	14.5	15.7	17.1	18.6	20.2	22.0	24.0	NULL	NULL	NULL	-1.1
35	NULL	NULL	NULL	NULL	NULL	NULL	9.0	10.0	11.1	12.2	13.3	14.5	15.6	17.0	18.5	20.1	22.0	NULL	NULL	NULL	1.7
40	NULL	NULL	NULL	NULL	NULL	NULL	8.9	9.9	10.9	11.9	13.0	14.1	15.3	16.8	18.0	19.8	21.8	NULL	NULL	NULL	4.4
45	NULL	NULL	NULL	NULL	NULL	NULL	8.8	9.8	10.8	11.7	12.8	13.9	15.1	16.7	17.9	19.7	21.7	NULL	NULL	NULL	7.2
50	NULL	NULL	NULL	NULL	NULL	NULL	8.7	9.7	10.7	11.7	12.7	13.8	15.0	16.6	17.8	19.6	21.6	NULL	NULL	NULL	10.0
55	NULL	NULL	NULL	NULL	NULL	NULL	8.6	9.7	10.7	11.6	12.6	13.7	14.9	16.5	17.7	19.5	21.3	NULL	NULL	NULL	12.8
60	NULL	NULL	NULL	NULL	NULL	NULL	8.5	9.6	10.6	11.6	12.5	13.6	14.8	16.4	17.5	19.5	21.1	NULL	NULL	NULL	15.6
65	NULL	NULL	NULL	NULL	NULL	NULL	8.4	9.4	10.4	11.5	12.4	13.5	14.7	16.3	17.3	19.4	20.8	NULL	NULL	NULL	18.3
70	NULL	NULL	NULL	NULL	NULL	NULL	8.2	9.2	10.2	11.1	12.2	13.3	14.5	15.6	17.0	18.6	20.4	NULL	NULL	NULL	21.1
75	NULL	NULL	NULL	NULL	NULL	NULL	8.0	8.9	9.9	10.9	11.8	12.9	14.0	15.2	16.6	18.1	19.9	NULL	NULL	NULL	23.9
80	NULL	NULL	NULL	NULL	NULL	NULL	7.9	8.9	9.7	10.8	11.7	12.8	13.8	15.0	16.4	17.8	19.6	NULL	NULL	NULL	26.7
85	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	29.4
90	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	32.2
95	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	35.0
100	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	37.8
Adapted from Source: https://pami.ca/wp-content/uploads/2024/01/NAD2BFactsheet_FINAL_2020-08-24-1.pdf																					
Downloaded 2024May3																					
Equation: Modified Henderson																					
(NULL: Out of Chart Range from Original Data Source)																					

Long Grain Rice

RICE (LONG GRAIN)																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-1.1
35	NULL	NULL	NULL	NULL	9.2	10.1	10.9	11.7	12.5	13.3	14.1	14.9	15.7	16.6	17.6	18.6	19.8	21.3	NULL	1.7
40	NULL	NULL	NULL	NULL	9.0	9.9	10.7	11.5	12.3	13	13.8	14.6	15.4	16.3	17.2	18.2	19.4	20.9	NULL	4.4
45	NULL	NULL	NULL	NULL	8.8	9.7	10.5	11.2	12	12.8	13.5	14.3	15.1	15.9	16.9	17.9	19.0	20.5	NULL	7.2
50	NULL	NULL	NULL	NULL	8.6	9.5	10.3	11.0	11.8	12.5	13.3	14	14.8	15.7	16.5	17.5	18.7	20.1	NULL	10.0
55	NULL	NULL	NULL	NULL	8.5	9.3	10.1	10.8	11.5	12.3	13	13.8	14.5	15.4	16.3	17.2	18.4	19.8	NULL	12.8
60	NULL	NULL	NULL	NULL	8.3	9.1	9.9	10.6	11.3	12.1	12.8	13.5	14.3	15.1	16	16.9	18.1	19.5	NULL	15.6
65	NULL	NULL	NULL	NULL	8.2	8.9	9.7	10.4	11.1	11.9	12.6	13.3	14.1	14.9	15.7	16.7	17.8	19.2	NULL	18.3
70	NULL	NULL	NULL	NULL	8.0	8.8	9.5	10.3	11	11.7	12.4	13.1	13.8	14.6	15.5	16.4	17.5	18.9	NULL	21.1
75	NULL	NULL	NULL	NULL	7.9	8.7	9.4	10.1	10.8	11.5	12.2	12.9	13.6	14.4	15.2	16.2	17.2	18.6	NULL	23.9
80	NULL	NULL	NULL	NULL	7.8	8.5	9.2	9.9	10.6	11.3	12	12.7	13.4	14.2	15	15.9	17.0	18.3	NULL	26.7
85	NULL	NULL	NULL	NULL	7.6	8.4	9.1	9.8	10.5	11.1	11.8	12.5	13.2	14	14.8	15.7	16.8	18.1	NULL	29.4
90	NULL	NULL	NULL	NULL	7.5	8.3	9	9.6	10.3	11	11.6	12.3	13	13.8	14.6	15.5	16.5	17.8	NULL	32.2
95	NULL	NULL	NULL	NULL	7.4	8.1	8.8	9.5	10.2	10.8	11.5	12.2	12.9	13.6	14.4	15.3	16.3	17.6	NULL	35.0
100	NULL	NULL	NULL	NULL	7.3	8	8.7	9.4	10	10.7	11.3	12	12.7	13.4	14.2	15.1	16.1	17.4	NULL	37.8
Data Source: https://www.uaex.uada.edu/farm-ranch/crops-commercial-horticulture/Grain_drying_and_storage/Docs/FSA_1074_Grain_Drying_Tools_Equilibrium_Moisture_Content_Tables_and_Psychrometric_Charts.pdf																				
Downloaded 2024May3																				
Equation: (Unknown)																				
(NULL: Out of Chart Range from Original Data Source)																				

Sorghum

SORGHUM																					
Calculated Equilibrium Moisture																					
Temperature(F)	Humidity(%)																				Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%		
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7	
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9	
30	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-1.1	
35	NULL	NULL	NULL	NULL	11.5	12.1	12.7	13.3	13.8	14.4	15	15.6	16.3	17.0	17.8	18.8	19.9	21.4	NULL	1.7	
40	NULL	NULL	NULL	NULL	11.3	11.9	12.5	13.1	13.6	14.2	14.8	15.4	16.1	16.8	17.6	18.6	19.7	21.2	NULL	4.4	
45	NULL	NULL	NULL	NULL	11.1	11.7	12.3	12.9	13.4	14	14.6	15.3	15.9	16.6	17.5	18.4	19.6	21.1	NULL	7.2	
50	NULL	NULL	NULL	NULL	10.9	11.5	12.1	12.7	13.3	13.8	14.4	15.1	15.7	16.5	17.3	18.2	19.4	20.9	NULL	10.0	
55	NULL	NULL	NULL	NULL	10.7	11.3	11.9	12.5	13.1	13.7	14.3	14.9	15.6	16.3	17.1	18.1	19.2	20.8	NULL	12.8	
60	NULL	NULL	NULL	NULL	10.5	11.2	11.7	12.3	12.9	13.5	14.1	14.7	15.4	16.1	17.0	17.9	19.1	20.6	NULL	15.6	
65	NULL	NULL	NULL	NULL	10.4	11	11.6	12.2	12.7	13.3	13.9	14.6	15.2	16.0	16.8	17.8	18.9	20.5	NULL	18.3	
70	NULL	NULL	NULL	NULL	10.2	10.8	11.4	12	12.6	13.2	13.8	14.4	15.1	15.8	16.7	17.6	18.8	20.3	NULL	21.1	
75	NULL	NULL	NULL	NULL	10.0	10.6	11.2	11.8	12.4	13	13.6	14.3	14.9	15.7	16.5	17.5	18.7	20.2	NULL	23.9	
80	NULL	NULL	NULL	NULL	9.9	10.5	11.1	11.7	12.3	12.9	13.5	14.1	14.8	15.6	16.4	17.4	18.5	20.1	NULL	26.7	
85	NULL	NULL	NULL	NULL	9.7	10.3	10.9	11.5	12.1	12.7	13.3	14.0	14.7	15.4	16.3	17.2	18.4	20.0	NULL	29.4	
90	NULL	NULL	NULL	NULL	9.6	10.2	10.8	11.4	12	12.6	13.2	13.8	14.5	15.3	16.1	17.1	18.3	19.8	NULL	32.2	
95	NULL	NULL	NULL	NULL	9.4	10	10.6	11.2	11.8	12.4	13.1	13.7	14.4	15.2	16.0	17.0	18.2	19.7	NULL	35.0	
100	NULL	NULL	NULL	NULL	9.3	9.9	10.5	11.1	11.7	12.3	12.9	13.6	14.3	15.0	15.9	16.9	18.0	19.6	NULL	37.8	
Data Source: https://www.uaex.uada.edu/farm-ranch/crops-commercial-horticulture/Grain_drying_and_storage/Docs/FSA_1074_Grain_Drying_Tools_Equilibrium_Moisture_Content_Tables_and_Psychrometric_Charts.pdf																					
Downloaded 2024May3																					
Equation: (Unkown)																					
(NULL: Out of Chart Range from Original Data Source)																					

SOYBEANS																					
Calculated Equilibrium Moisture																					
Temperature(F)	Humidity(%)																			Temperature(C)	
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%		
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7	
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9	
30	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-1.1	
35	NULL	NULL	NULL	NULL	5.9	6.5	7.1	7.8	8.6	9.4	10.3	11.5	12.8	14.4	16.4	19.1	22.9	28.9	NULL	1.7	
40	NULL	NULL	NULL	NULL	5.8	6.4	7.1	7.7	8.5	9.3	10.2	11.3	12.6	14.2	16.2	18.9	22.7	28.7	NULL	4.4	
45	NULL	NULL	NULL	NULL	5.8	6.4	7.0	7.7	8.4	9.2	10.1	11.2	12.5	14.1	16.1	18.7	22.5	28.4	NULL	7.2	
50	NULL	NULL	NULL	NULL	5.7	6.3	6.9	7.6	8.3	9.1	10.0	11.1	12.4	14.0	16.0	18.6	22.3	28.2	NULL	10.0	
55	NULL	NULL	NULL	NULL	5.7	6.2	6.8	7.5	8.2	9.0	10.0	11.0	12.3	13.8	15.8	18.4	22.1	28.0	NULL	12.8	
60	NULL	NULL	NULL	NULL	5.6	6.2	6.8	7.4	8.1	8.9	9.9	10.9	12.2	13.7	15.7	18.3	21.9	27.8	NULL	15.6	
65	NULL	NULL	NULL	NULL	5.6	6.1	6.7	7.4	8.1	8.9	9.8	10.8	12.1	13.6	15.5	18.1	21.7	27.6	NULL	18.3	
70	NULL	NULL	NULL	NULL	5.5	6.1	6.6	7.3	8.0	8.8	9.7	10.7	11.9	13.5	15.4	17.9	21.6	27.3	NULL	21.1	
75	NULL	NULL	NULL	NULL	5.4	6.0	6.6	7.2	7.9	8.7	9.6	10.6	11.8	13.3	15.2	17.8	21.4	27.1	NULL	23.9	
80	NULL	NULL	NULL	NULL	5.4	5.9	6.5	7.1	7.8	8.6	9.5	10.5	11.7	13.2	15.1	17.6	21.2	26.9	NULL	26.7	
85	NULL	NULL	NULL	NULL	5.3	5.9	6.4	7.1	7.7	8.5	9.4	10.4	11.6	13.1	15.0	17.5	21.0	26.7	NULL	29.4	
90	NULL	NULL	NULL	NULL	5.3	5.8	6.4	7.0	7.7	8.4	9.3	10.3	11.5	13.0	14.8	17.3	20.8	26.5	NULL	32.2	
95	NULL	NULL	NULL	NULL	5.2	5.7	6.3	6.9	7.6	8.3	9.2	10.2	11.4	12.8	14.7	17.1	20.7	26.3	NULL	35.0	
100	NULL	NULL	NULL	NULL	5.2	5.7	6.2	6.9	7.5	8.3	9.1	10.1	11.3	12.7	14.5	17.0	20.5	26.1	NULL	37.8	
Data Source: https://www.uaex.uada.edu/farm-ranch/crops-commercial-horticulture/Grain_drying_and_storage/Docs/FSA_1074_Grain_Drying_Tools_Equilibrium_Moisture_Content_Tables_and_Psychrometric_Charts.pdf																					
Downloaded 2024May3																					
Equation: (Unkown)																					
(NULL: Out of Chart Range from Original Data Source)																					

(NULL: Out of Chart Range from Original Data Source)

Confectionery Sunflower

CONFECTIONERY SUNFLOWER																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	2.8	4.1	5.1	6.0	6.9	7.7	8.4	9.2	9.9	10.7	11.4	12.2	13.0	13.9	14.8	15.9	17.1	18.7	21.0	-6.7
25	2.7	4.0	5.0	5.9	6.7	7.4	8.2	8.9	9.6	10.4	11.1	11.9	12.7	13.5	14.4	15.5	16.7	18.2	20.4	-3.9
30	2.6	3.8	4.8	5.7	6.5	7.2	7.9	8.7	9.4	10.1	10.8	11.5	12.3	13.1	14.0	15.1	16.2	17.7	19.9	-1.1
35	2.6	3.7	4.7	5.5	6.3	7.0	7.7	8.4	9.1	9.8	10.5	11.2	12.0	12.8	13.7	14.7	15.8	17.3	19.5	1.7
40	2.5	3.6	4.6	5.4	6.1	6.9	7.5	8.2	8.9	9.6	10.2	11.0	11.7	12.5	13.4	14.3	15.5	16.9	19.0	4.4
45	2.4	3.5	4.5	5.3	6.0	6.7	7.4	8.0	8.7	9.3	10.0	10.7	11.4	12.2	13.1	14.0	15.1	16.5	18.6	7.2
50	2.4	3.5	4.3	5.1	5.8	6.5	7.2	7.8	8.5	9.1	9.8	10.5	11.2	11.9	12.8	13.7	14.8	16.2	18.2	10.0
55	2.3	3.4	4.2	5.0	5.7	6.4	7.0	7.7	8.3	8.9	9.6	10.2	10.9	11.7	12.5	13.4	14.5	15.9	17.9	12.8
60	2.3	3.3	4.2	4.9	5.6	6.2	6.9	7.5	8.1	8.7	9.4	10.0	10.7	11.5	12.3	13.2	14.2	15.6	17.5	15.6
65	2.2	3.2	4.1	4.8	5.5	6.1	6.7	7.3	7.9	8.6	9.2	9.8	10.5	11.2	12.0	12.9	14.0	15.3	17.2	18.3
70	2.2	3.2	4.0	4.7	5.4	6.0	6.6	7.2	7.8	8.4	9.0	9.6	10.3	11.0	11.8	12.7	13.7	15.0	16.9	21.1
75	2.1	3.1	3.9	4.6	5.3	5.9	6.5	7.1	7.6	8.2	8.8	9.5	10.1	10.8	11.6	12.4	13.5	14.7	16.6	23.9
80	2.1	3.0	3.8	4.5	5.2	5.8	6.4	6.9	7.5	8.1	8.7	9.3	9.9	10.6	11.4	12.2	13.2	14.5	16.4	26.7
85	2.0	3.0	3.8	4.4	5.1	5.7	6.2	6.8	7.4	7.9	8.5	9.1	9.8	10.4	11.2	12.0	13.0	14.3	16.1	29.4
90	2.0	2.9	3.7	4.4	5.0	5.6	6.1	6.7	7.2	7.8	8.4	9.0	9.6	10.3	11.0	11.8	12.8	14.0	15.8	32.2
95	2.0	2.9	3.6	4.3	4.9	5.5	6.0	6.6	7.1	7.7	8.2	8.8	9.5	10.1	10.8	11.6	12.6	13.8	15.6	35.0
100	1.9	2.8	3.6	4.2	4.8	5.4	5.9	6.5	7.0	7.6	8.1	8.7	9.3	10.0	10.7	11.5	12.4	13.6	15.4	37.8

Download Source : <https://www.ag.ndsu.edu/graindrying/documents/GrainEquilibriumMoistureContentCharts.pdf>

Equation: ASAE Paper 74-3534 (Unknown Equation)

Downloaded 2024May3

Dr. Kenneth J Hellevang, PE

NDSU Extension Service

Fargo, ND 58105

Sunflower Hull

SUNFLOWER (HULLS)																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	NULL	9.7	10.5	11.2	12.0	13.0	14.2	15.4	16.9	18.8	21.2	24.6	NULL	-1.1
35	NULL	NULL	NULL	NULL	NULL	NULL	8.6	9.1	9.9	10.6	11.5	12.5	13.7	15.0	16.7	18.9	21.9	NULL	NULL	1.7
40	NULL	NULL	NULL	NULL	NULL	NULL	8.1	8.8	9.5	10.1	11.0	11.9	13.0	14.3	15.9	18.0	21.0	NULL	NULL	4.4
45	NULL	NULL	NULL	NULL	NULL	NULL	7.8	8.4	9.0	9.7	10.5	11.6	12.6	13.8	15.4	17.6	20.5	NULL	NULL	7.2
50	NULL	NULL	NULL	NULL	NULL	NULL	7.8	8.2	9.0	9.7	10.5	11.4	12.5	13.5	15.3	17.4	20.2	NULL	NULL	10.0
55	NULL	NULL	NULL	NULL	NULL	NULL	7.4	7.9	8.6	9.4	10.0	10.9	11.9	13.1	14.7	16.6	19.4	NULL	NULL	12.8
60	NULL	NULL	NULL	NULL	NULL	NULL	7.4	7.9	8.5	9.3	10.0	10.9	11.9	13.1	14.6	16.6	19.4	NULL	NULL	15.6
65	NULL	NULL	NULL	NULL	NULL	NULL	7.1	7.6	8.2	8.8	9.5	10.4	11.3	12.5	13.9	15.8	18.6	NULL	NULL	18.3
70	NULL	NULL	NULL	NULL	NULL	NULL	6.5	6.9	7.4	8.0	8.7	9.5	10.4	11.5	12.8	14.6	17.1	NULL	NULL	21.1
75	NULL	NULL	NULL	NULL	NULL	NULL	5.9	6.4	6.9	7.4	8.1	8.8	9.6	10.6	11.9	13.5	15.8	NULL	NULL	23.9
80	NULL	NULL	NULL	NULL	NULL	NULL	5.8	6.3	6.8	7.4	8.0	8.7	9.5	10.5	11.7	13.4	15.7	NULL	NULL	26.7
85	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	29.4
90	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	32.2
95	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	35.0
100	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	37.8
Adapted from Source: https://pami.ca/wp-content/uploads/2021/10/Equilibrium-Moisture-Content-Charts-for-Grain-Storage-Management_rev2.pdf																				
Downloaded 2024May3																				
Equation: Modified Halsey																				
(NULL: Out of Chart Range from Original Data Source)																				

Sunflower Kernel

SUNFLOWER (KERNELS)																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	4.5	4.9	5.2	5.6	6.0	6.5	7.2	7.8	8.8	9.9	11.6	NULL	NULL	-1.1
35	NULL	NULL	NULL	NULL	NULL	NULL	4.0	4.3	4.7	5.0	5.4	5.9	6.5	7.1	7.9	8.9	10.4	NULL	NULL	1.7
40	NULL	NULL	NULL	NULL	NULL	NULL	3.9	4.2	4.6	4.9	5.3	5.8	6.2	6.8	7.7	8.7	10.1	NULL	NULL	4.4
45	NULL	NULL	NULL	NULL	NULL	NULL	3.9	4.1	4.5	4.8	5.1	5.5	6.1	6.7	7.5	8.4	10.0	NULL	NULL	7.2
50	NULL	NULL	NULL	NULL	NULL	NULL	3.9	4.2	4.5	4.8	5.2	5.7	6.2	6.8	7.6	8.6	10.0	NULL	NULL	10.0
55	NULL	NULL	NULL	NULL	NULL	NULL	3.8	4.1	4.4	4.7	5.1	5.5	6.0	6.6	7.3	8.3	9.7	NULL	NULL	12.8
60	NULL	NULL	NULL	NULL	NULL	NULL	3.8	4.1	4.4	4.7	5.1	5.5	6.0	6.6	7.4	8.4	9.9	NULL	NULL	15.6
65	NULL	NULL	NULL	NULL	NULL	NULL	3.7	3.9	4.2	4.5	4.9	5.3	5.9	6.5	7.2	8.2	9.6	NULL	NULL	18.3
70	NULL	NULL	NULL	NULL	NULL	NULL	3.4	3.7	4.0	4.3	4.7	5.1	5.5	6.1	6.7	7.7	9.0	NULL	NULL	21.1
75	NULL	NULL	NULL	NULL	NULL	NULL	3.2	3.5	3.8	4.1	4.4	4.8	5.2	5.7	6.4	7.3	8.5	NULL	NULL	23.9
80	NULL	NULL	NULL	NULL	NULL	NULL	3.3	3.5	3.8	4.1	4.4	4.8	5.2	5.7	6.4	7.2	8.5	NULL	NULL	26.7
85	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	29.4
90	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	32.2
95	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	35.0
100	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	37.8
Adapted from Source: https://pami.ca/wp-content/uploads/2021/10/Equilibrium-Moisture-Content-Charts-for-Grain-Storage-Management_rev2.pdf																				
Downloaded 2024May3																				
Equation: Modified Halsey																				
(NULL: Out of Chart Range from Original Data Source)																				

Sunflower (Oil Hybrids)

SUNFLOWER (OIL HYBRIDS)																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	2.7	3.7	4.5	5.1	5.6	6.2	6.7	7.1	7.6	8.1	8.6	9.0	9.5	10.1	10.6	11.3	12.0	12.9	14.3	-6.7
25	2.7	3.6	4.3	5.0	5.5	6.0	6.5	7.0	7.4	7.9	8.3	8.8	9.3	9.8	10.4	11.0	11.7	12.6	14.0	-3.9
30	2.6	3.5	4.2	4.8	5.4	5.9	6.4	6.8	7.3	7.7	8.2	8.6	9.1	9.6	10.2	10.8	11.5	12.4	13.7	-1.1
35	2.5	3.5	4.1	4.7	5.3	5.7	6.2	6.7	7.1	7.5	8.0	8.4	8.9	9.4	9.9	10.5	11.2	12.1	13.4	1.7
40	2.5	3.4	4.1	4.6	5.1	5.6	6.1	6.5	7.0	7.4	7.8	8.3	8.7	9.2	9.7	10.3	11.0	11.9	13.1	4.4
45	2.4	3.3	4.0	4.5	5.0	5.5	6.0	6.4	6.8	7.2	7.7	8.1	8.6	9.0	9.6	10.1	10.8	11.7	12.9	7.2
50	2.4	3.2	3.9	4.5	4.9	5.4	5.8	6.3	6.7	7.1	7.5	8.0	8.4	8.9	9.4	10.0	10.6	11.5	12.7	10.0
55	2.3	3.2	3.8	4.4	4.9	5.3	5.7	6.2	6.6	7.0	7.4	7.8	8.3	8.7	9.2	9.8	10.4	11.3	12.5	12.8
60	2.3	3.1	3.8	4.3	4.8	5.2	5.6	6.1	6.5	6.9	7.3	7.7	8.1	8.6	9.1	9.6	10.3	11.1	12.3	15.6
65	2.3	3.1	3.7	4.2	4.7	5.1	5.5	6.0	6.3	6.7	7.1	7.6	8.0	8.4	8.9	9.5	10.1	10.9	12.1	18.3
70	2.2	3.0	3.6	4.2	4.6	5.0	5.5	5.9	6.2	6.6	7.0	7.4	7.9	8.3	8.8	9.3	9.9	10.7	11.9	21.1
75	2.2	3.0	3.6	4.1	4.5	5.0	5.4	5.8	6.2	6.5	6.9	7.3	7.7	8.2	8.7	9.2	9.8	10.6	11.7	23.9
80	2.2	2.9	3.5	4.0	4.5	4.9	5.3	5.7	6.1	6.4	6.8	7.2	7.6	8.1	8.5	9.0	9.7	10.4	11.5	26.7
85	2.1	2.9	3.5	4.0	4.4	4.8	5.2	5.6	6.0	6.3	6.7	7.1	7.5	7.9	8.4	8.9	9.5	10.3	11.4	29.4
90	2.1	2.8	3.4	3.9	4.3	4.8	5.1	5.5	5.9	6.3	6.6	7.0	7.4	7.8	8.3	8.8	9.4	10.1	11.2	32.2
95	2.1	2.8	3.4	3.9	4.3	4.7	5.1	5.4	5.8	6.2	6.5	6.9	7.3	7.7	8.2	8.7	9.3	10.0	11.1	35.0
100	2.0	2.8	3.3	3.8	4.2	4.6	5.0	5.4	5.7	6.1	6.5	6.8	7.2	7.6	8.1	8.6	9.2	9.9	11.0	37.8

Download Source : <https://www.ag.ndsu.edu/graindrying/documents/GrainEquilibriumMoistureContentCharts.pdf>

Equation: ASAE Paper 74-3534 (Unknown Equation)

Downloaded 2024May3

Dr. Kenneth J Hellevang, PE

NDSU Extension Service

Fargo, ND 58105

Sunflower Seed

SUNFLOWER (SEEDS)																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	NULL	5.8	6.2	6.7	7.1	8.3	9.0	9.9	11.0	12.4	14.4	NULL	NULL	-1.1
35	NULL	NULL	NULL	NULL	NULL	NULL	NULL	5.2	5.6	6.0	6.5	7.0	7.5	8.2	8.9	9.9	11.2	13.1	NULL	1.7
40	NULL	NULL	NULL	NULL	NULL	NULL	NULL	5.1	5.5	5.9	6.2	6.7	7.3	7.9	8.7	9.7	10.9	12.7	NULL	4.4
45	NULL	NULL	NULL	NULL	NULL	NULL	NULL	5.0	5.3	5.8	6.2	6.7	7.2	7.9	8.6	9.6	10.8	12.6	NULL	7.2
50	NULL	NULL	NULL	NULL	NULL	NULL	NULL	5.0	5.4	5.7	6.2	6.6	7.1	7.8	8.4	9.4	10.7	12.4	NULL	10.0
55	NULL	NULL	NULL	NULL	NULL	NULL	NULL	4.9	5.3	5.7	6.1	6.5	7.1	7.7	8.4	9.4	10.5	12.3	NULL	12.8
60	NULL	NULL	NULL	NULL	NULL	NULL	NULL	4.9	5.3	5.6	6.1	6.5	7.0	7.6	8.3	9.3	10.6	12.2	NULL	15.6
65	NULL	NULL	NULL	NULL	NULL	NULL	NULL	4.7	5.1	5.5	5.9	6.4	6.9	7.5	8.2	9.1	10.3	12.0	NULL	18.3
70	NULL	NULL	NULL	NULL	NULL	NULL	NULL	4.5	4.8	5.2	5.6	6.0	6.5	7.0	7.7	8.6	9.8	11.3	NULL	21.1
75	NULL	NULL	NULL	NULL	NULL	NULL	NULL	4.3	4.6	5.0	5.2	5.7	6.1	6.7	7.4	8.2	9.3	10.8	NULL	23.9
80	NULL	NULL	NULL	NULL	NULL	NULL	NULL	4.3	4.6	4.9	5.2	5.6	6.1	6.7	7.4	8.2	9.2	10.8	NULL	26.7
85	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	29.4
90	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	32.2
95	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	35.0
100	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	37.8
Adapted from Source: https://pami.ca/wp-content/uploads/2021/10/Equilibrium-Moisture-Content-Charts-for-Grain-Storage-Management_rev2.pdf																				
Downloaded 2024May3																				
Equation: Modified Halsey																				
(NULL: Out of Chart Range from Original Data Source)																				

Wheat (Durum)

WHEAT (DURUM)																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																		Temperature(C)	
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%		95%
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-1.1
35	NULL	NULL	NULL	NULL	8.3	8.9	9.6	10.2	10.9	11.5	12.2	13.0	13.8	14.7	15.8	17.1	18.8	21.3	NULL	1.7
40	NULL	NULL	NULL	NULL	8.2	8.8	9.5	10.1	10.7	11.4	12.1	12.8	13.7	14.6	15.7	17.0	18.7	21.1	NULL	4.4
45	NULL	NULL	NULL	NULL	8.1	8.8	9.4	10.0	10.6	11.3	12.0	12.7	13.5	14.4	15.5	16.8	18.5	20.9	NULL	7.2
50	NULL	NULL	NULL	NULL	8	8.7	9.3	9.9	10.5	11.2	11.9	12.6	13.4	14.3	15.4	16.6	18.3	20.7	NULL	10.0
55	NULL	NULL	NULL	NULL	7.9	8.6	9.2	9.8	10.4	11.1	11.7	12.5	13.3	14.2	15.2	16.5	18.1	20.5	NULL	12.8
60	NULL	NULL	NULL	NULL	7.8	8.5	9.1	9.7	10.3	10.9	11.6	12.3	13.1	14.0	15.1	16.3	18.0	20.4	NULL	15.6
65	NULL	NULL	NULL	NULL	7.8	8.4	9.0	9.6	10.2	10.8	11.5	12.2	13.0	13.9	14.9	16.2	17.8	20.2	NULL	18.3
70	NULL	NULL	NULL	NULL	7.7	8.3	8.9	9.5	10.1	10.7	11.4	12.1	12.8	13.7	14.7	16.0	17.6	20.0	NULL	21.1
75	NULL	NULL	NULL	NULL	7.6	8.2	8.8	9.4	10.0	10.6	11.2	11.9	12.7	13.6	14.6	15.8	17.4	19.8	NULL	23.9
80	NULL	NULL	NULL	NULL	7.5	8.1	8.7	9.3	9.9	10.5	11.1	11.8	12.6	13.4	14.4	15.7	17.3	19.6	NULL	26.7
85	NULL	NULL	NULL	NULL	7.4	8.0	8.6	9.2	9.8	10.4	11.0	11.7	12.4	13.3	14.3	15.5	17.1	19.4	NULL	29.4
90	NULL	NULL	NULL	NULL	7.3	7.9	8.5	9.1	9.6	10.2	10.9	11.5	12.3	13.1	14.1	15.3	16.9	19.2	NULL	32.2
95	NULL	NULL	NULL	NULL	7.2	7.8	8.4	9.0	9.5	10.1	10.7	11.4	12.2	13.0	14.0	15.2	16.7	19.0	NULL	35.0
100	NULL	NULL	NULL	NULL	7.2	7.7	8.3	8.8	9.4	10.0	10.6	11.3	12.0	12.8	13.8	15.0	16.5	18.8	NULL	37.8
Data Source: https://www.uaex.uada.edu/farm-ranch/crops-commercial-horticulture/Grain_drying_and_storage/Docs/FSA 1074 Grain Drying Tools Equilibrium Moisture Content Tables and Psychrometric Charts.pdf																				
Downloaded 2024May3																				
Equation: (Unknown)																				
(NULL: Out of Chart Range from Original Data Source)																				

Wheat (Hard)

WHEAT (HARD)																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-1.1
35	NULL	NULL	NULL	NULL	9.5	10.4	11.2	11.9	12.7	13.4	14.1	14.9	15.7	16.5	17.3	18.3	19.4	20.7	NULL	1.7
40	NULL	NULL	NULL	NULL	9.4	10.2	11.0	11.7	12.4	13.2	13.9	14.6	15.4	16.2	17.0	18.0	19.1	20.4	NULL	4.4
45	NULL	NULL	NULL	NULL	9.2	10.0	10.8	11.5	12.2	12.9	13.7	14.4	15.1	15.9	16.8	17.7	18.8	20.1	NULL	7.2
50	NULL	NULL	NULL	NULL	9.0	9.8	10.6	11.3	12.0	12.7	13.4	14.2	14.9	15.7	16.5	17.4	18.5	19.8	NULL	10.0
55	NULL	NULL	NULL	NULL	8.9	9.7	10.4	11.1	11.8	12.5	13.2	13.9	14.7	15.4	16.3	17.2	18.2	19.5	NULL	12.8
60	NULL	NULL	NULL	NULL	8.7	9.5	10.3	11.0	11.7	12.3	13.0	13.7	14.4	15.2	16.0	16.9	17.9	19.2	NULL	15.6
65	NULL	NULL	NULL	NULL	8.6	9.4	10.1	10.8	11.5	12.2	12.8	13.5	14.2	15.0	15.8	16.7	17.7	19.0	NULL	18.3
70	NULL	NULL	NULL	NULL	8.5	9.2	10.0	10.7	11.3	12.0	12.7	13.3	14.0	14.8	15.6	16.5	17.5	18.7	NULL	21.1
75	NULL	NULL	NULL	NULL	8.4	9.1	9.8	10.5	11.2	11.8	12.5	13.2	13.9	14.6	15.4	16.2	17.2	18.5	NULL	23.9
80	NULL	NULL	NULL	NULL	8.3	9.0	9.7	10.4	11.0	11.7	12.3	13.0	13.7	14.4	15.2	16.0	17.0	18.3	NULL	26.7
85	NULL	NULL	NULL	NULL	8.1	8.9	9.6	10.2	10.9	11.5	12.2	12.8	13.5	14.2	15.0	15.8	16.8	18.1	NULL	29.4
90	NULL	NULL	NULL	NULL	8.0	8.8	9.4	10.1	10.7	11.4	12.0	12.7	13.3	14.1	14.8	15.7	16.6	17.8	NULL	32.2
95	NULL	NULL	NULL	NULL	7.9	8.7	9.3	10.0	10.6	11.2	11.9	12.5	13.2	13.9	14.6	15.5	16.4	17.6	NULL	35.0
100	NULL	NULL	NULL	NULL	7.8	8.6	9.2	9.9	10.5	11.1	11.7	12.4	13.0	13.7	14.5	15.3	16.3	17.5	NULL	37.8
Download Source : https://www.ag.ndsu.edu/graindrying/documents/GrainEquilibriumMoistureContentCharts.pdf																				
Equation: ASAE 245.4 Modified Henderson																				
Downloaded 2024May3																				
Dr. Kenneth J Hellevang, PE																				
NDSU Extension Service																				
Fargo, ND 58105																				

Wheat (Soft)

WHEAT (SOFT)																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-1.1
35	NULL	NULL	NULL	NULL	9.0	9.8	10.4	11.1	11.7	12.3	12.9	13.5	14.2	14.8	15.5	16.3	17.2	18.3	NULL	1.7
40	NULL	NULL	NULL	NULL	8.9	9.6	10.3	10.9	11.5	12.1	12.7	13.3	14.0	14.6	15.3	16.1	17.0	18.1	NULL	4.4
45	NULL	NULL	NULL	NULL	8.8	9.5	10.1	10.8	11.4	12.0	12.6	13.2	13.8	14.4	15.1	15.9	16.8	17.9	NULL	7.2
50	NULL	NULL	NULL	NULL	8.7	9.4	10.0	10.6	11.2	11.8	12.4	13.0	13.6	14.2	14.9	15.7	16.6	17.6	NULL	10.0
55	NULL	NULL	NULL	NULL	8.6	9.2	9.9	10.5	11.1	11.7	12.2	12.8	13.4	14.1	14.8	15.5	16.4	17.4	NULL	12.8
60	NULL	NULL	NULL	NULL	8.4	9.1	9.8	10.4	10.9	11.5	12.1	12.7	13.3	13.9	14.6	15.3	16.2	17.2	NULL	15.6
65	NULL	NULL	NULL	NULL	8.3	9.0	9.6	10.2	10.8	11.4	12.0	12.5	13.1	13.7	14.4	15.2	16.0	17.0	NULL	18.3
70	NULL	NULL	NULL	NULL	8.2	8.9	9.5	10.1	10.7	11.3	11.8	12.4	13.0	13.6	14.3	15.0	15.8	16.9	NULL	21.1
75	NULL	NULL	NULL	NULL	8.2	8.8	9.4	10.0	10.6	11.1	11.7	12.3	12.8	13.4	14.1	14.8	15.7	16.7	NULL	23.9
80	NULL	NULL	NULL	NULL	8.1	8.7	9.3	9.9	10.5	11.0	11.6	12.1	12.7	13.3	14.0	14.7	15.5	16.5	NULL	26.7
85	NULL	NULL	NULL	NULL	8.0	8.6	9.2	9.8	10.3	10.9	11.4	12.0	12.6	13.2	13.8	14.5	15.3	16.4	NULL	29.4
90	NULL	NULL	NULL	NULL	7.9	8.5	9.1	9.7	10.2	10.8	11.3	11.9	12.4	13.0	13.7	14.4	15.2	16.2	NULL	32.2
95	NULL	NULL	NULL	NULL	7.8	8.4	9.0	9.6	10.1	10.7	11.2	11.8	12.3	12.9	13.6	14.3	15.1	16.1	NULL	35.0
100	NULL	NULL	NULL	NULL	7.7	8.4	8.9	9.5	10.0	10.6	11.1	11.7	12.2	12.8	13.4	14.1	14.9	15.9	NULL	37.8

Download Source : <https://www.ag.ndsu.edu/graindrying/documents/GrainEquilibriumMoistureContentCharts.pdf>

Equation: ASAE 245.4 Modified Henderson

Downloaded 2024May3

Dr. Kenneth J Hellevang, PE

NDSU Extension Service

Fargo, ND 58105

Wheat (Hard Red Spring)

WHEAT (HARD RED SPRING)																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	11.3	12.0	12.7	13.4	14.3	15.0	15.8	16.8	17.9	19.0	20.4	NULL	NULL	-1.1
35	NULL	NULL	NULL	NULL	NULL	NULL	10.2	10.9	11.6	12.3	13.1	13.8	14.6	15.4	16.4	17.5	18.8	NULL	NULL	1.7
40	NULL	NULL	NULL	NULL	NULL	NULL	10.0	10.7	11.4	12.1	12.9	13.6	14.4	15.3	16.3	17.4	18.5	NULL	NULL	4.4
45	NULL	NULL	NULL	NULL	NULL	NULL	9.9	10.6	11.3	12.0	12.8	13.5	14.3	15.1	16.0	17.2	18.4	NULL	NULL	7.2
50	NULL	NULL	NULL	NULL	NULL	NULL	9.9	10.6	11.3	11.9	12.8	13.5	14.3	15.1	15.9	17.0	18.4	NULL	NULL	10.0
55	NULL	NULL	NULL	NULL	NULL	NULL	9.8	10.5	11.2	11.8	12.6	13.4	14.2	15.0	15.8	17.0	18.3	NULL	NULL	12.8
60	NULL	NULL	NULL	NULL	NULL	NULL	9.7	10.5	11.2	11.6	12.6	13.4	14.1	14.9	15.8	16.8	18.3	NULL	NULL	15.6
65	NULL	NULL	NULL	NULL	NULL	NULL	9.6	10.4	11.1	11.5	12.4	13.3	14.0	14.7	15.7	16.7	18.1	NULL	NULL	18.3
70	NULL	NULL	NULL	NULL	NULL	NULL	9.2	10.0	10.7	11.3	12.0	12.8	13.6	14.7	15.4	16.5	17.9	NULL	NULL	21.1
75	NULL	NULL	NULL	NULL	NULL	NULL	8.9	9.6	10.3	11.0	11.7	12.4	13.3	14.0	15.1	16.2	17.6	NULL	NULL	23.9
80	NULL	NULL	NULL	NULL	NULL	NULL	8.9	9.5	10.2	10.9	11.6	12.4	13.3	13.9	14.9	16.1	17.4	NULL	NULL	26.7
85	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	29.4
90	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	32.2
95	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	35.0
100	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	37.8
Adapted from Source: https://pami.ca/wp-content/uploads/2021/10/Equilibrium-Moisture-Content-Charts-for-Grain-Storage-Management_rev2.pdf																				
Downloaded 2024May3																				
Equation: Modified Chung-Pfost																				
(NULL: Out of Chart Range from Original Data Source)																				

Wheat (Hard Red Winter)

WHEAT (HARD RED WINTER)																				
Calculated Equilibrium Moisture																				
Temperature(F)	Humidity(%)																			Temperature(C)
	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%	90%	95%	
20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-6.7
25	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	-3.9
30	NULL	NULL	NULL	NULL	NULL	NULL	NULL	10.8	11.8	13.0	14.2	15.4	16.8	18.4	20.2	22.3	24.9	28.4	NULL	-1.1
35	NULL	NULL	NULL	NULL	NULL	NULL	NULL	9.8	10.8	11.8	13.0	14.1	15.3	16.8	18.5	20.4	22.8	26.1	NULL	1.7
40	NULL	NULL	NULL	NULL	NULL	NULL	NULL	9.7	10.6	11.6	12.7	13.8	15.1	16.6	18.0	20.0	22.4	25.7	NULL	4.4
45	NULL	NULL	NULL	NULL	NULL	NULL	NULL	9.6	10.5	11.5	12.5	13.6	15.0	16.5	17.8	19.7	22.3	25.5	NULL	7.2
50	NULL	NULL	NULL	NULL	NULL	NULL	NULL	9.5	10.4	11.4	12.5	13.6	14.9	16.4	17.6	19.6	22.1	25.4	NULL	10.0
55	NULL	NULL	NULL	NULL	NULL	NULL	NULL	9.4	10.3	11.3	12.4	13.5	14.8	16.2	17.4	19.4	22.0	25.2	NULL	12.8
60	NULL	NULL	NULL	NULL	NULL	NULL	NULL	9.4	10.3	11.3	12.3	13.5	14.7	16.1	17.1	19.1	21.7	25.0	NULL	15.6
65	NULL	NULL	NULL	NULL	NULL	NULL	NULL	9.3	10.2	11.1	12.1	13.4	14.6	16.0	16.9	18.9	21.5	24.6	NULL	18.3
70	NULL	NULL	NULL	NULL	NULL	NULL	NULL	8.8	9.8	10.7	11.6	12.7	13.9	15.3	16.8	18.6	20.8	23.9	NULL	21.1
75	NULL	NULL	NULL	NULL	NULL	NULL	NULL	8.5	9.4	10.3	11.4	12.2	13.4	14.8	16.2	18.3	20.1	23.0	NULL	23.9
80	NULL	NULL	NULL	NULL	NULL	NULL	NULL	8.5	9.4	10.3	11.3	12.2	13.3	14.7	16.0	17.9	19.8	22.8	NULL	26.7
85	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	29.4
90	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	32.2
95	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	35.0
100	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	37.8
Adapted from Source: https://pami.ca/wp-content/uploads/2021/10/Equilibrium-Moisture-Content-Charts-for-Grain-Storage-Management_rev2.pdf																				
Downloaded 2024May3																				
Equation: Modified Oswin																				
(NULL: Out of Chart Range from Original Data Source)																				

Quick Reference Guide

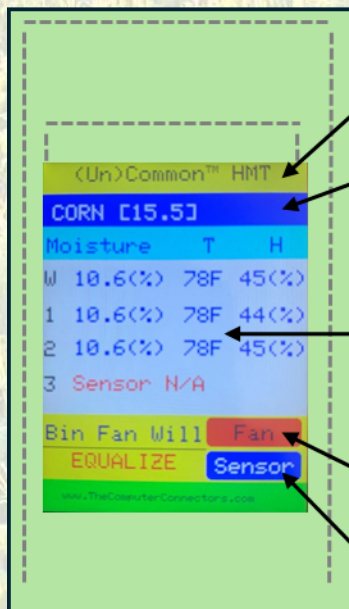
Powering the Device

NOTE: We STRONGLY advise you to use the USB port to power this device. The other power connections on the circuit board are NOT adequately protected.

- **ALWAYS use the USB connection!**
- Wall Charger or Portable Battery



The Dashboard Screen



• Set your PRO Version license codes by touching here.

• Set the Grain Type by touching here.

• Switch between Fahrenheit and Celsius by touching anywhere in the sensor data area.

• Touching in this area will also force an immediate sensor read.

• Set the fan operation limits by touching here.

• Set the sensor calibration values by touching here.

A Word From Our Sponsor

Thank you for your interest in this project. Even though (or perhaps because) a lot of time and money went into the research and development, it is very rewarding to see it come to fruition.

We hope that you had some fun building this device. We certainly did!

Tell all your friends and neighbors!

Please let us know about any challenges or ideas you have. We are always looking for ways to improve. As such, we do plan on occasionally releasing new firmware updates. And, in a lot of cases, those firmware updates will add new features. And of course, the firmware updates are FREE. So, visit our website (www.TheComputerConnectors.com) often.

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